

Module 2 – Advanced Group Policy Administration

Task 1 – Create and Link a Group Policy Object (GPO)

Objective

Create and link a new Group Policy Object (GPO) to the **Workstations** Organizational Unit (OU) to prepare for centralized management of workstation settings.

Background

Group Policy allows administrators to centrally configure and manage Windows computers and users within an Active Directory domain. By linking a GPO to an Organizational Unit, administrators can apply consistent security settings, desktop configurations, software deployments, and other policies to all objects contained within that OU.

Systems Used

Component	Value
Server	DC01
Operating System	Windows Server 2022
Domain	benlab.local
Management Tool	Group Policy Management Console (GPMC)
Target OU	Workstations

Procedure

1. Opened **Group Policy Management** (gpmc.msc).
 2. Expanded the **benlab.local** domain.
 3. Navigated to **BEN Industries** → **Workstations**.
 4. Right-clicked the **Workstations** OU.
 5. Selected **Create a GPO in this domain, and Link it here...**
 6. Created a new GPO named **Corporate Desktop**.
 7. Verified that the GPO was successfully linked to the **Workstations** OU.
-

Result

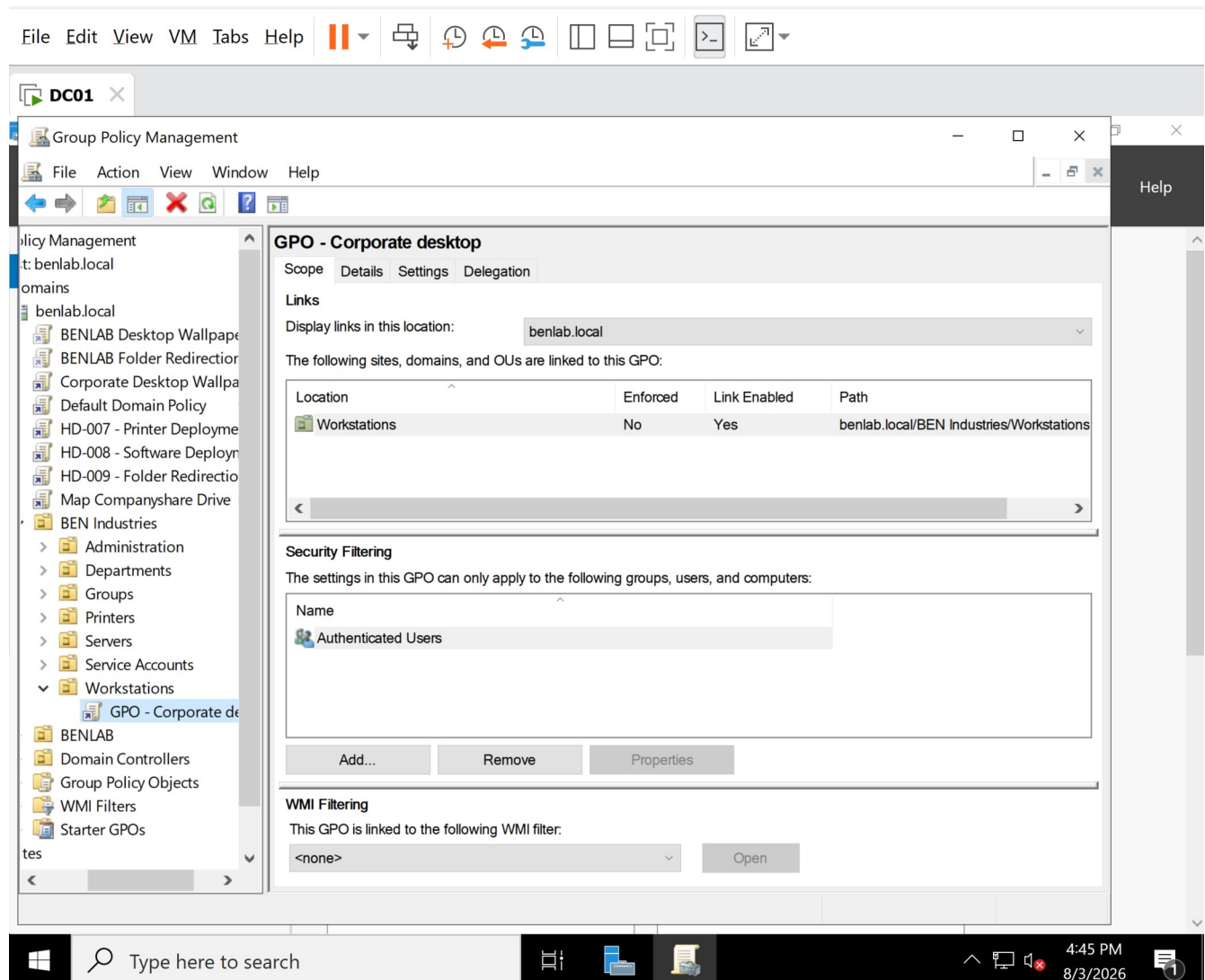
A new Group Policy Object named **Corporate Desktop** was successfully created and linked to the **Workstations** Organizational Unit. The GPO is now ready to be configured with administrative policies for Windows client computers.

Verification

Verified in **Group Policy Management Console** that:

- The **Corporate Desktop** GPO exists.
- The GPO is linked to the **Workstations** OU.
- The link is enabled and available for future policy configuration.

Screenshot



Corporate desktop GPO linked to the Workstation organization Unit

Skills Demonstrated

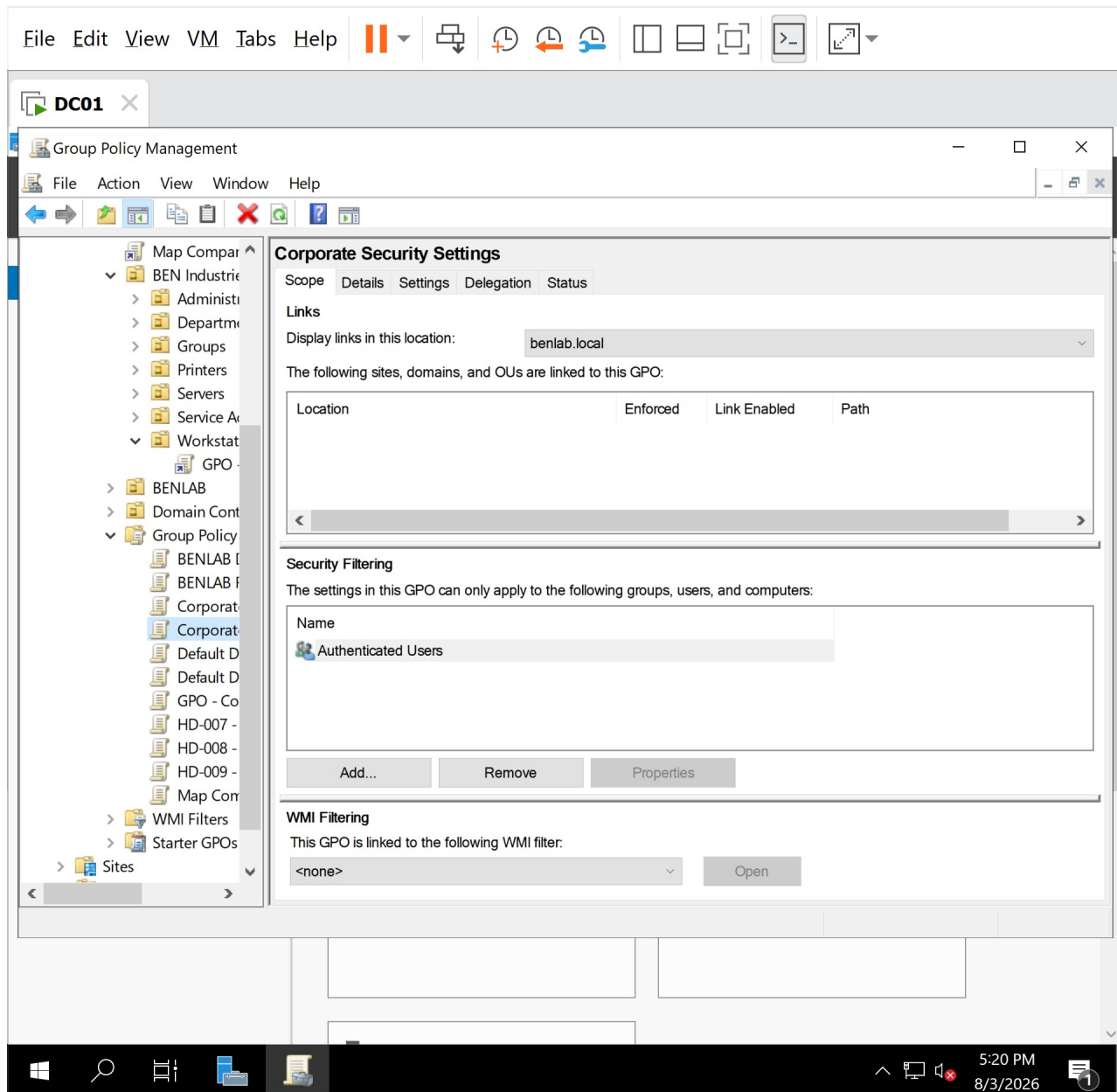
- Group Policy Management
- Organizational Unit Administration
- Active Directory Administration
- Enterprise Windows Infrastructure
- Centralized Policy Management

Task 2 - Documentation

- **Objective**
 - Create a dedicated Group Policy Object for enterprise security settings.
- **Background**
 - Explain why administrators create separate GPOs instead of modifying the Default Domain Policy. Mention that separating policies makes management, troubleshooting, and future expansion easier.
- **Systems Used**
 - DC01
 - Windows Server 2022
 - Domain: benlab.local
- **Steps Performed**
 1. Opened **Group Policy Management**.
 2. Expanded **Forest** → **Domains** → **benlab.local** → **Group Policy Objects**.
 3. Right-clicked **Group Policy Objects**.
 4. Selected **New**.
 5. Named the GPO **Corporate Security Settings**.
 6. Verified the GPO was created successfully.
- **Result**
 - Successfully created a new GPO that will later contain enterprise security policies.

Screenshot

Screenshots



Task 3 - Configure Enterprise Password Policy Using Group Policy

Objective

Configure a centralized enterprise password policy using Group Policy to enforce strong password requirements for domain users. This task demonstrates how administrators implement organization-wide password policies to improve security and ensure compliance with enterprise security standards.

Background

Group Policy enables administrators to centrally manage password policies across an Active Directory domain. Strong password policies help protect user accounts from unauthorized access, password reuse, and brute-force attacks by enforcing consistent security requirements throughout the organization.

Unlike Fine-Grained Password Policies, which target specific users or security groups, Group Policy is commonly used to establish the organization's standard password policy.

Systems Used

Component	Value
Domain Controller	DC01
Operating System	Windows Server 2022
Domain	benlab.local
Management Tool	Group Policy Management Console (GPMC)
Group Policy Object	Corporate Security Settings

Procedure

Step 1

Opened **Group Policy Management** (gpmc.msc).

Expanded:

Forest: benlab.local

Domains

benlab.local

Group Policy Objects

Corporate Security Settings

Right-clicked **Corporate Security Settings** and selected **Edit**.

Step 2

Navigated to:

Computer Configuration

Policies

Windows Settings

Step 3

Configured the following enterprise password policy settings.

Policy	Configured Value
Enforce password history	24 passwords remembered
Maximum password age	90 days
Minimum password age	1 day
Minimum password length	12 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Explanation of Each Policy

Enforce Password History

Configured to remember the previous **24 passwords**.

This prevents users from immediately reusing previously used passwords and encourages stronger password rotation.

Maximum Password Age

Configured to **90 days**.

Users are required to change their passwords every 90 days, reducing the likelihood of compromised passwords remaining active for extended periods.

Minimum Password Age

Configured to **1 day**.

This prevents users from rapidly changing their password multiple times in succession to bypass the password history requirement while still allowing legitimate password changes the following day if necessary.

Minimum Password Length

Configured to **12 characters**.

Longer passwords provide significantly stronger protection against password guessing and brute-force attacks than shorter passwords.

Password Must Meet Complexity Requirements

Configured as **Enabled**.

Passwords must include a combination of character types such as uppercase letters, lowercase letters, numbers, and special characters, reducing the risk of weak or easily guessed passwords.

Store Passwords Using Reversible Encryption

Configured as **Disabled**.

This setting remains disabled because storing passwords using reversible encryption significantly reduces password security and is generally unnecessary in modern enterprise environments.

Verification

Verified that all password policy settings were successfully configured within the **Corporate Security Settings** Group Policy Object.

Confirmed that the configured values appeared correctly in the Password Policy console.

Captured screenshots documenting each stage of the configuration.

Comparison with Fine-Grained Password Policies

This task demonstrates configuring an organization-wide password policy using **Group Policy**.

Earlier in **Module 1**, a **Fine-Grained Password Policy (Password Settings Object - PSO)** was created.

Group Policy

- Applies the organization's standard password policy.
- Used for broad password management across domain users.

Fine-Grained Password Policy

- Applies different password requirements to specific users or security groups.
- Commonly used for privileged accounts such as Domain Administrators, Finance Administrators, and IT Administrators.

Using both technologies together provides flexibility while maintaining a consistent enterprise security strategy.

Results

Successfully created and configured an enterprise password policy using Group Policy.

The **Corporate Security Settings** GPO now enforces standardized password security requirements that can be applied to domain-joined computers and users after the GPO is linked and deployed.

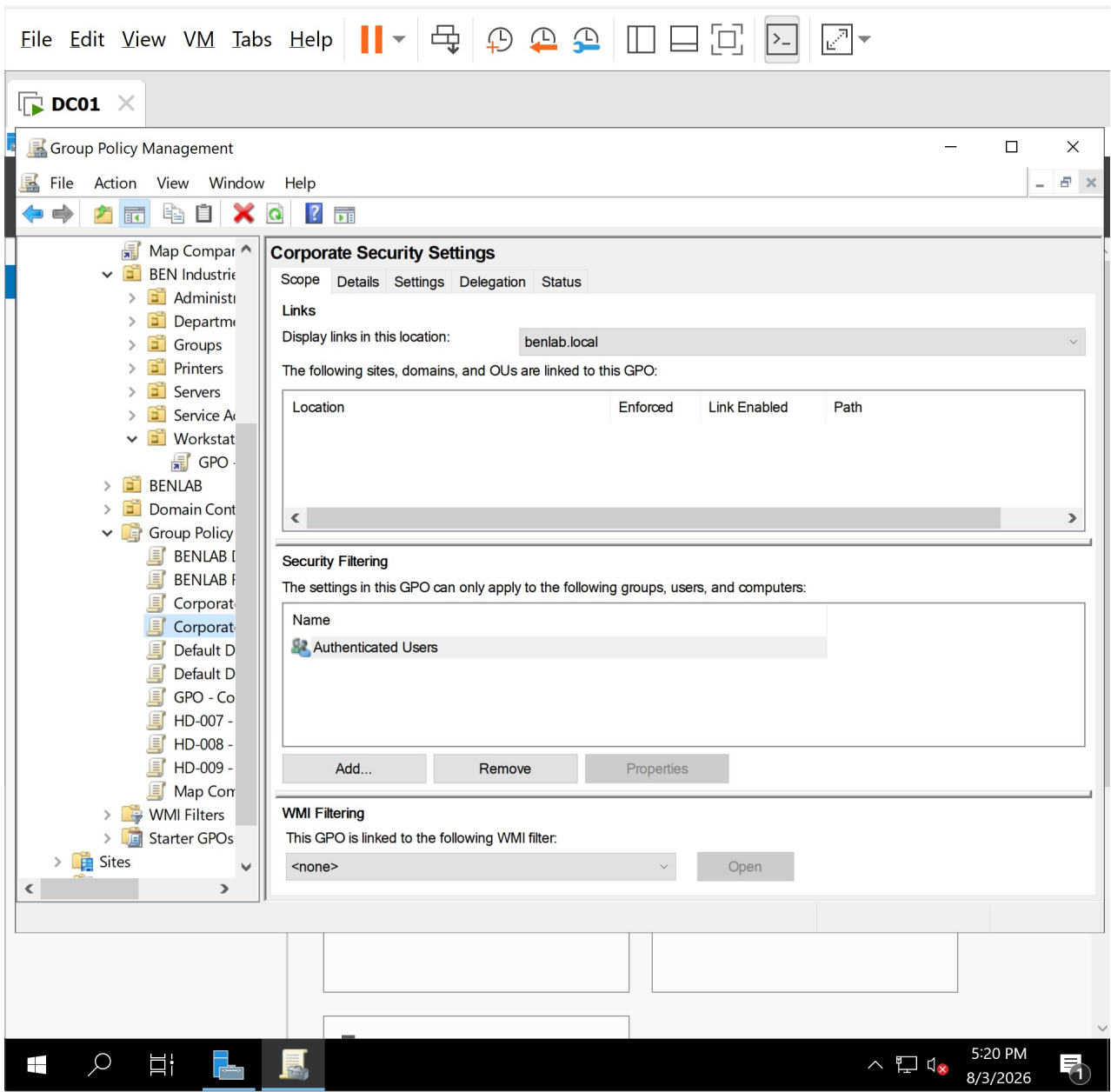
Skills Demonstrated

- Windows Server Administration
 - Active Directory Administration
 - Group Policy Management
 - Enterprise Password Policy Configuration
 - Security Hardening
 - Enterprise Authentication Management
 - Windows Security Administration
-

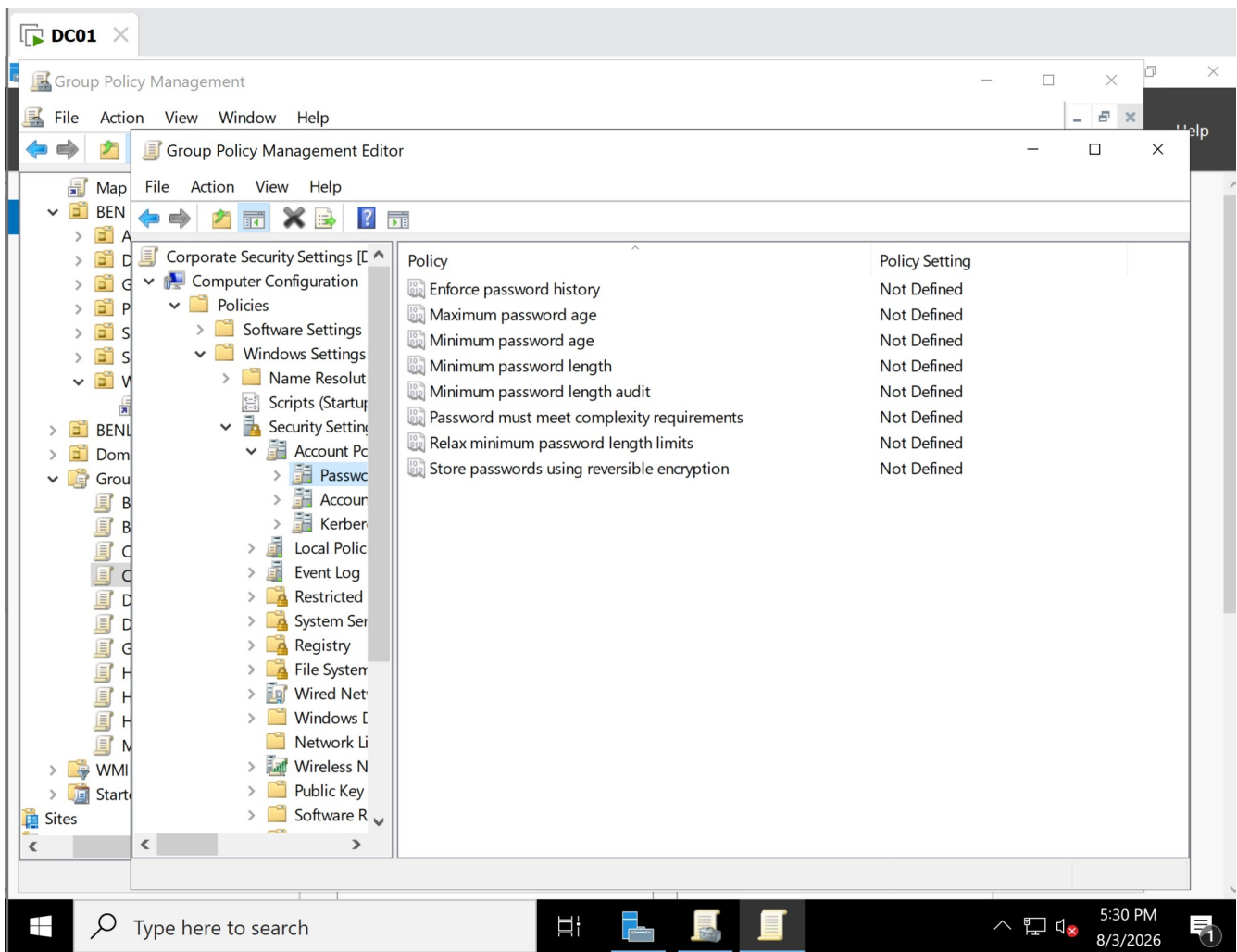
Lessons Learned

This exercise demonstrated how Group Policy provides centralized management of enterprise password policies. Understanding the relationship between Group Policy and Fine-Grained Password Policies is essential for Windows administrators because organizations often require a standard password policy for most users while applying stricter password requirements to privileged accounts through Password Settings Objects (PSOs). Configuring password policies through Group Policy ensures consistency, strengthens account security, and supports enterprise security best practices.

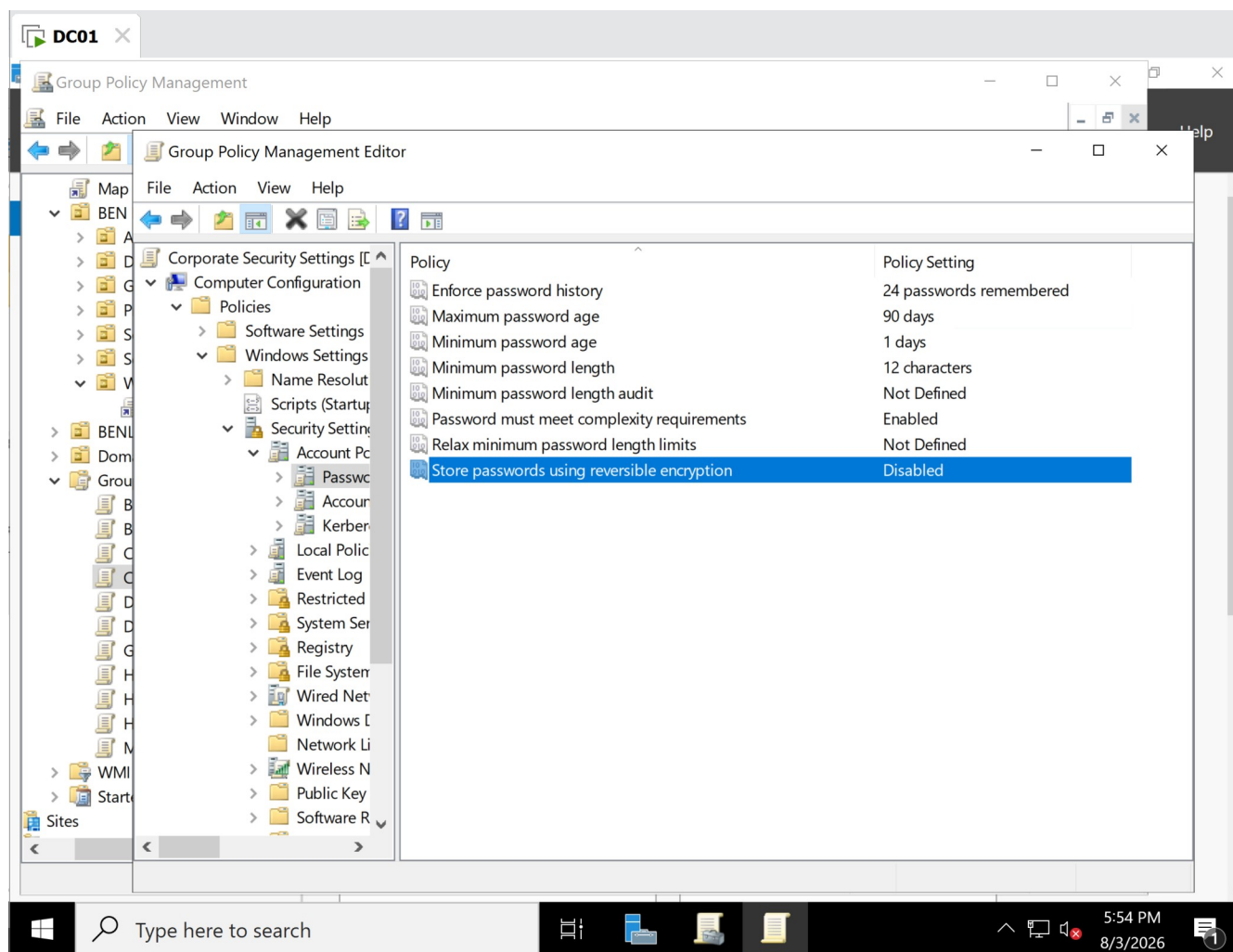
Screenshots



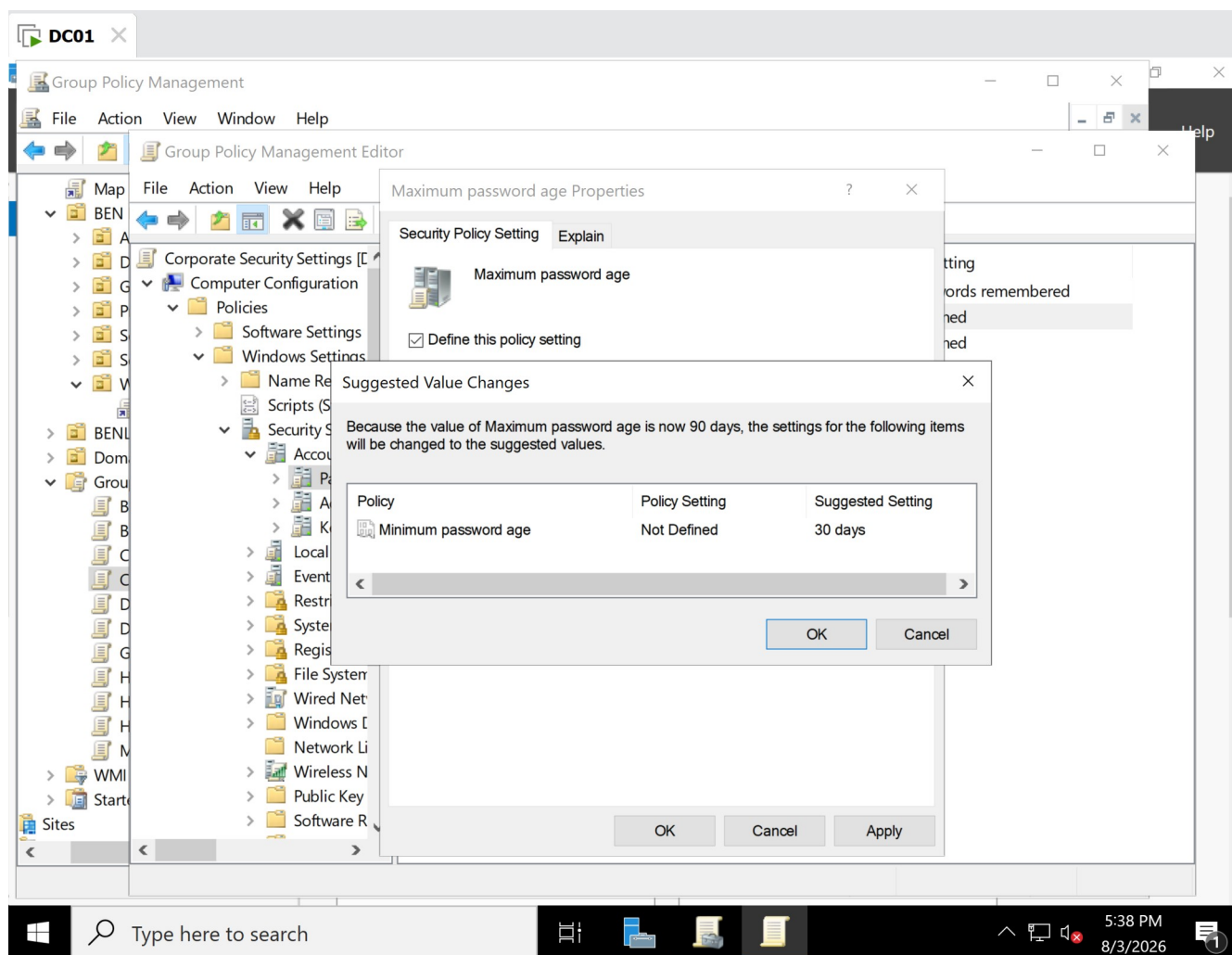
Corporate Security Settings GPO opened in Group Policy Management.



Password Policy node before configuration.



Enterprise password policy configured with organization security settings.



Example password policy configuration window (Maximum Password Age or Minimum Password Length).

Task 4 – Configure Account Lockout Policy

Objective

Configure an Account Lockout Policy using Group Policy to protect Active Directory user accounts against brute-force and password guessing attacks.

Background

Account Lockout Policies automatically lock user accounts after a specified number of failed logon attempts. This reduces the risk of unauthorized access by slowing automated password attacks while allowing administrators to balance security with user convenience.

Systems Used

Component	Value
Server	DC01
Operating System	Windows Server 2022
Domain	benlab.local
Tool	Group Policy Management Editor

Configuration Steps

1. Opened **Group Policy Management**.
2. Edited the **Corporate Security Settings** GPO.
3. Navigated to:

Computer Configuration

└ Policies
 └ Windows Settings
 └ Security Settings
 └ Account Policies
 └ Account Lockout Policy

4. Configured:
 - Account Lockout Threshold
 - Account Lockout Duration
 - Reset Account Lockout Counter After
5. Applied the changes.
6. Updated Group Policy.
7. Verified the effective account policy.

Commands Used

`gpupdate /force`

`net accounts`

Verification

The `net accounts` command confirmed the domain account policy, including the configured account lockout threshold and current lockout settings.

Result

Successfully configured and verified an Account Lockout Policy using Group Policy to strengthen Active Directory account security against repeated failed authentication attempts.

Task 5 - Configure Software Restriction Policies

Objective

Configure a Software Restriction Policy (SRP) through Group Policy to prevent users from running unauthorized applications from a specific folder. This demonstrates how Windows administrators can reduce security risks by restricting software execution.

Environment

- Domain Controller: DC01
 - Client Computer: CLIENT02
 - Domain: benlab.local
 - GPO: Corporate Security Settings
-

Procedure

1. Open the Group Policy Management Console

On DC01:

Server Manager → **Tools** → **Group Policy Management**

Navigate to:

Forest: benlab.local

Domains

benlab.local

BEN Industries

Workstations

Right-click **Corporate Security Settings** and select:

Edit

2. Create Software Restriction Policies

Navigate to:

Computer Configuration

Policies

Windows Settings

Security Settings

Software Restriction Policies

If Software Restriction Policies do not exist:

Action

Create New Policies

Windows automatically creates:

- Security Levels
- Additional Rules
- Enforcement
- Designated File Types
- Trusted Publishers

3. Create a Path Rule

Open:

Additional Rules

Right-click:

New Path Rule

Configure:

Path

C:\Games*

Security Level

Disallowed

Description

Block unauthorized games from executing.

Click:

OK

4. Update Group Policy

On CLIENT02, open Command Prompt as Administrator and run:

gpupdate /force

5. Verify GPO Application

Run:

```
gpresult /scope computer /r
```

Verification:

Applied Group Policy Objects

Corporate Security Settings

Default Domain Policy

HD-008 - Software Deployment

This confirmed that the **Corporate Security Settings** GPO was successfully applied to the workstation.

6. Test the Restriction

Create the folder:

```
mkdir C:\Games
```

Copy Command Prompt into the folder:

```
copy C:\Windows\System32\cmd.exe C:\Games\
```

Double-click:

C:\Games\cmd.exe

Windows displayed:

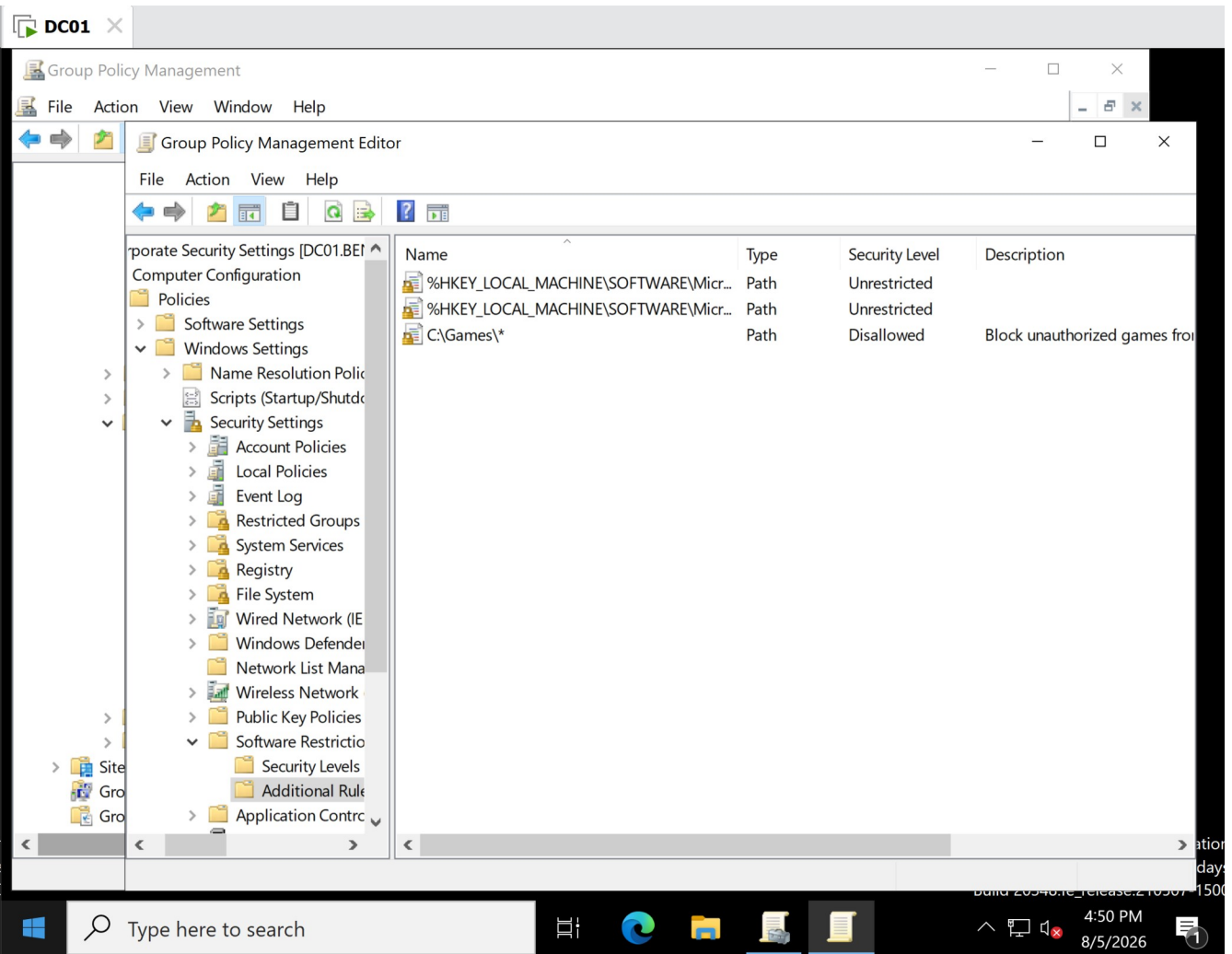
This app has been blocked by your system administrator.

This confirmed that the Software Restriction Policy successfully prevented executables from running from the restricted path.

Result

The Software Restriction Policy was successfully deployed through Group Policy and verified on CLIENT02. Executables located in the restricted folder (C:\Games) were blocked from running, demonstrating centralized application control using Windows Server Group Policy.

Screenshots



Software Restriction Policies → Additional Rules showing:

C:\Games*

Security Level: Disallowed

```
Home X Metasploitable2-Linux X CLIENT02 X SECONION01 X FILESRV01 X
Administrator: Command Prompt
RecyC:\Windows\System32>gpresult /scope computer /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on 8/ 5/ 2026 at 6:12:42 PM

RSOP data for on DESKTOP-0499MOA : Logging Mode
-----

OS Configuration:      Member Workstation
OS Version:            10.0.26200
Site Name:             Default-First-Site-Name
Roaming Profile:
Local Profile:
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=DESKTOP-0499MOA,OU=Workstations,OU=BEN Industries,DC=benlab,DC=local
Last time Group Policy was applied: 8/5/2026 at 5:47:17 PM
Group Policy was applied from:    DC01.benlab.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     BENLAB
Domain Type:                     Windows 2008 or later

Applied Group Policy Objects
-----
Corporate Security Settings
Default Domain Policy
HD-008 - Software Deployment

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The computer is a part of the following security groups
-----
BUILTIN\Administrators
Everyone
BUILTIN\Users
```

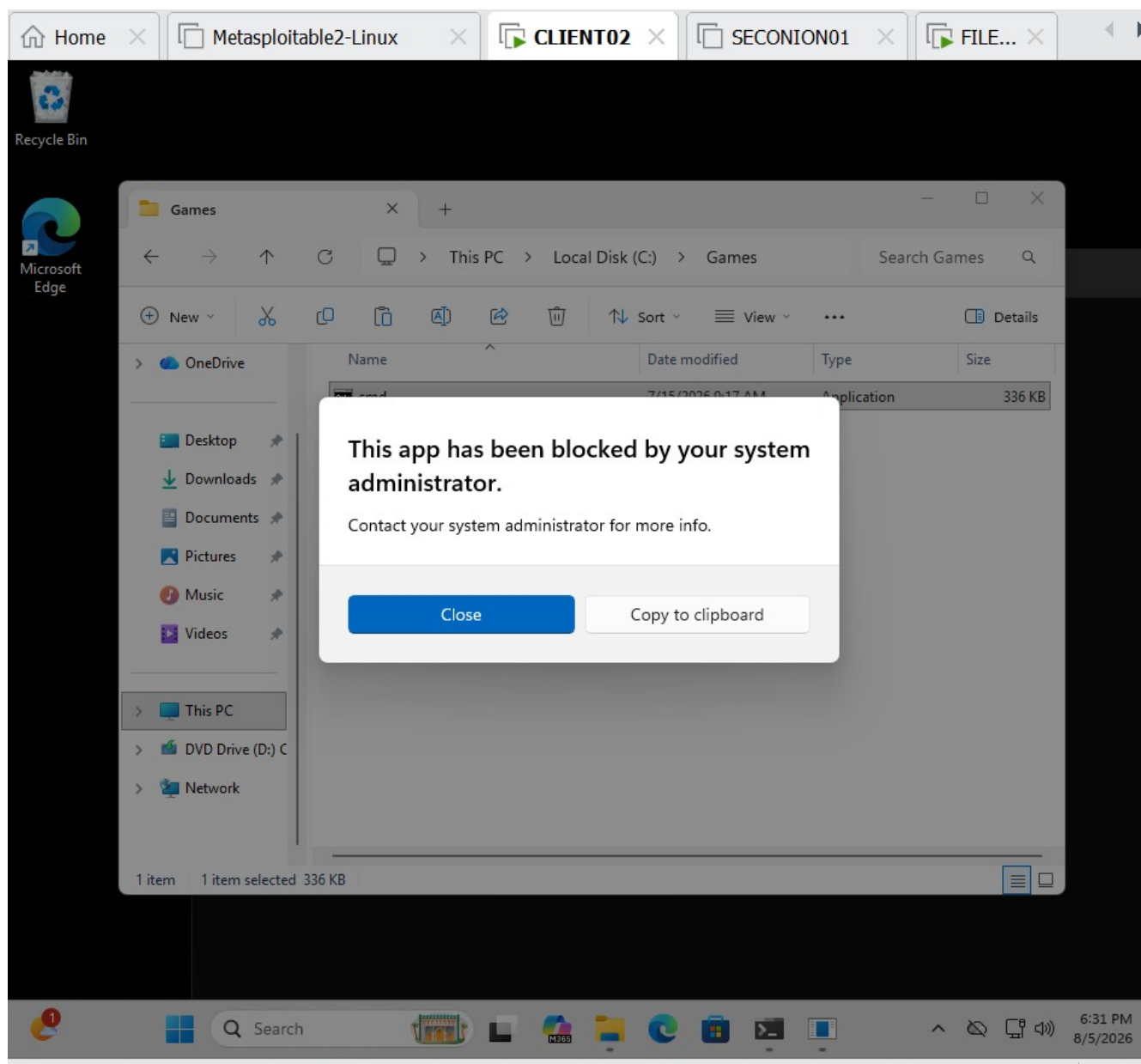
Output of:

gpresult /scope computer /r

showing:

Corporate Security Settings

under **Applied Group Policy Objects**.



The Windows message:

This app has been blocked by your system administrator.

Skills Demonstrated

- Group Policy Management
- Software Restriction Policies (SRP)
- Path Rules
- Application Whitelisting/Blacklisting
- Enterprise Endpoint Security

- Windows Server Administration
- Group Policy Verification (gpresult)
- Windows Security Hardening

Task 5 - Administrative Templates

Objective

Configure and verify Administrative Template settings using Group Policy to manage user access to Windows features in an Active Directory environment.

Environment

- **Domain Controller:** DC01 (Windows Server 2022)
- **Client Computer:** CLIENT02 (Windows 11)
- **Domain:** benlab.local
- **Test User:** Brian Johnson
- **Tool:** Group Policy Management Console (GPMC)

Configuration Steps

Step 1 – Edit the Existing GPO

1. Open **Group Policy Management** on DC01.
2. Navigate to:
Forest → Domains → benlab.local → Group Policy Objects
3. Right-click **Workstation Configuration**.
4. Select **Edit**.

Step 2 – Configure the Administrative Template

Navigate to:

User Configuration

└ Policies

└ Administrative Templates

└─ Control Panel

Enable the following policy:

- **Prohibit access to Control Panel and PC settings** → **Enabled**

Click **Apply**, then **OK**.

Step 3 – Link the GPO

Initially, the GPO was linked only to the **Workstations** Organizational Unit (OU). Because the configured setting was a **User Configuration** policy, it did not apply to users located in other OUs.

To ensure the policy applied to domain users, an additional link was created.

Steps performed:

1. Right-click the **Departments** OU.
2. Select **Link an Existing GPO**.
3. Choose **Workstation Configuration**.
4. Click **OK**.

The GPO was now linked to both:

- Workstations OU
 - Departments OU
-

Step 4 – Update Group Policy

On CLIENT02, logged in as **Brian Johnson**, run:

```
gpupdate /force
```

Step 5 – Verify the Applied Policies

Run:

```
gpresult /scope user /r
```

Verification confirmed the following GPOs were applied:

- Workstation Configuration
- Map CompanyShare Drive
- HD-007 – Printer Deployment
- HD-009 – Folder Redirection
- Corporate Desktop Wallpaper

Step 6 – Test the Policy

Attempt to open **Control Panel** while logged in as Brian Johnson.

Expected Result:

This operation has been cancelled due to restrictions in effect on this computer.
Please contact your system administrator.

The restriction confirmed that the Administrative Template policy was successfully applied.

Commands Used

```
gpupdate /force
```

```
gpresult /scope user /r
```

Verification

- ✓ Workstation Configuration GPO edited successfully.
 - ✓ Administrative Template policy configured.
 - ✓ GPO linked to both the Workstations and Departments OUs.
 - ✓ Group Policy successfully refreshed.
 - ✓ `gpresult` confirmed the GPO was applied.
 - ✓ Brian Johnson was unable to open Control Panel.
-

Troubleshooting Performed

During testing, the Administrative Template policy did not initially apply because the GPO was linked only to the **Workstations** OU. Since Administrative Template settings were configured under **User Configuration**, Windows evaluated the user's Organizational Unit instead of the computer's.

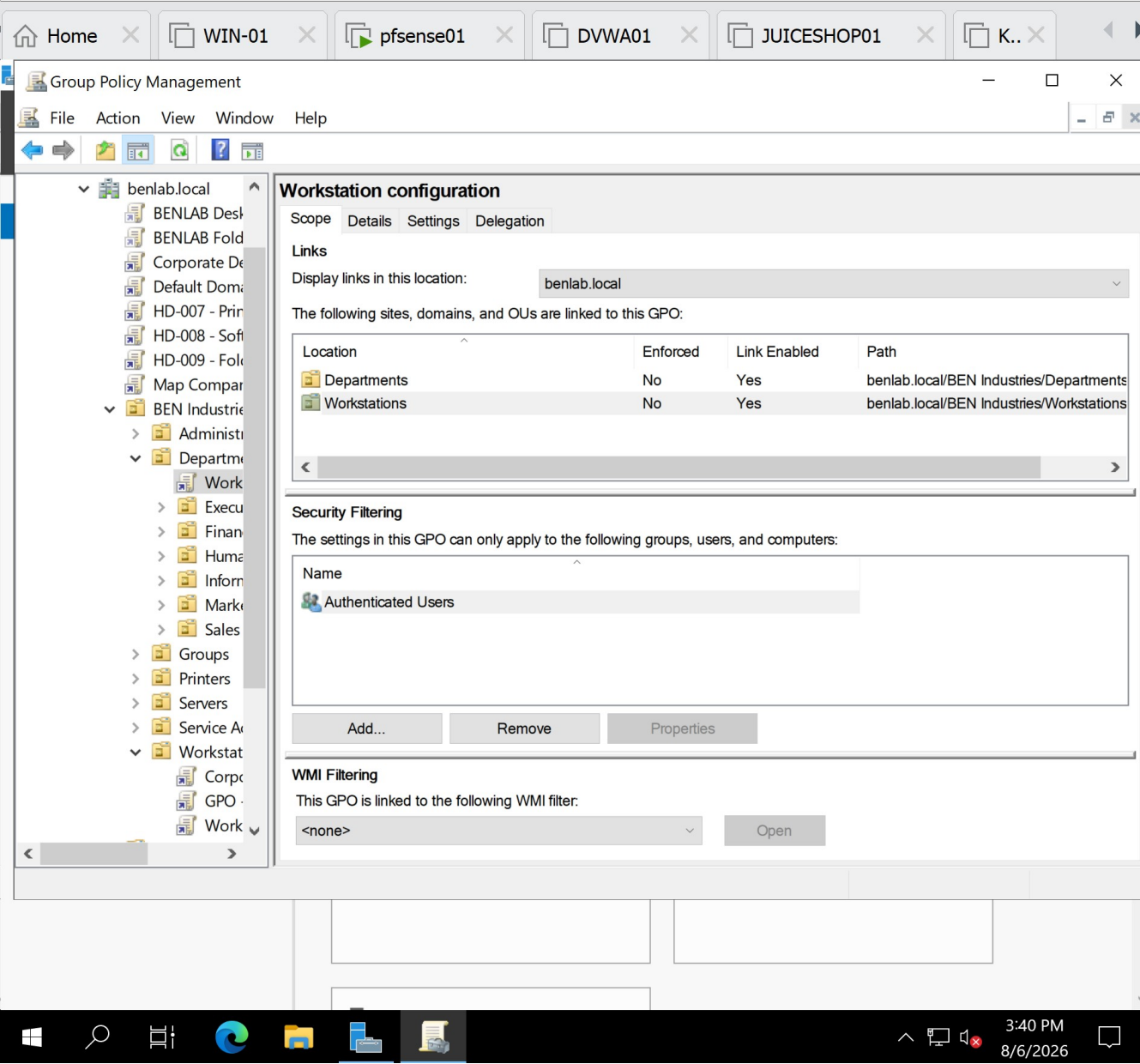
After linking the same GPO to the **Departments** OU, Group Policy was refreshed using `gpupdate /force`, and verification with `gpresult` confirmed that the policy was successfully applied to the user.

Skills Demonstrated

- Active Directory Group Policy Management

- Administrative Templates
 - Group Policy Linking
 - Organizational Unit (OU) Design
 - User Configuration Policies
 - Group Policy Troubleshooting
 - gpupdate
 - gpresult
 - Enterprise Windows Administration
-

Screenshots



GPO linked to the **Workstations** OU.

```
Command Prompt
USER SETTINGS
-----
CN=Brian Johnson,OU=Sales,OU=Departments,OU=BEN Industries,DC=benlab,DC=local
Last time Group Policy was applied: 8/6/2026 at 3:21:52 PM
Group Policy was applied from: DC01.benlab.local
Group Policy slow link threshold: 500 kbps
Domain Name: BENLAB
Domain Type: Windows 2008 or later

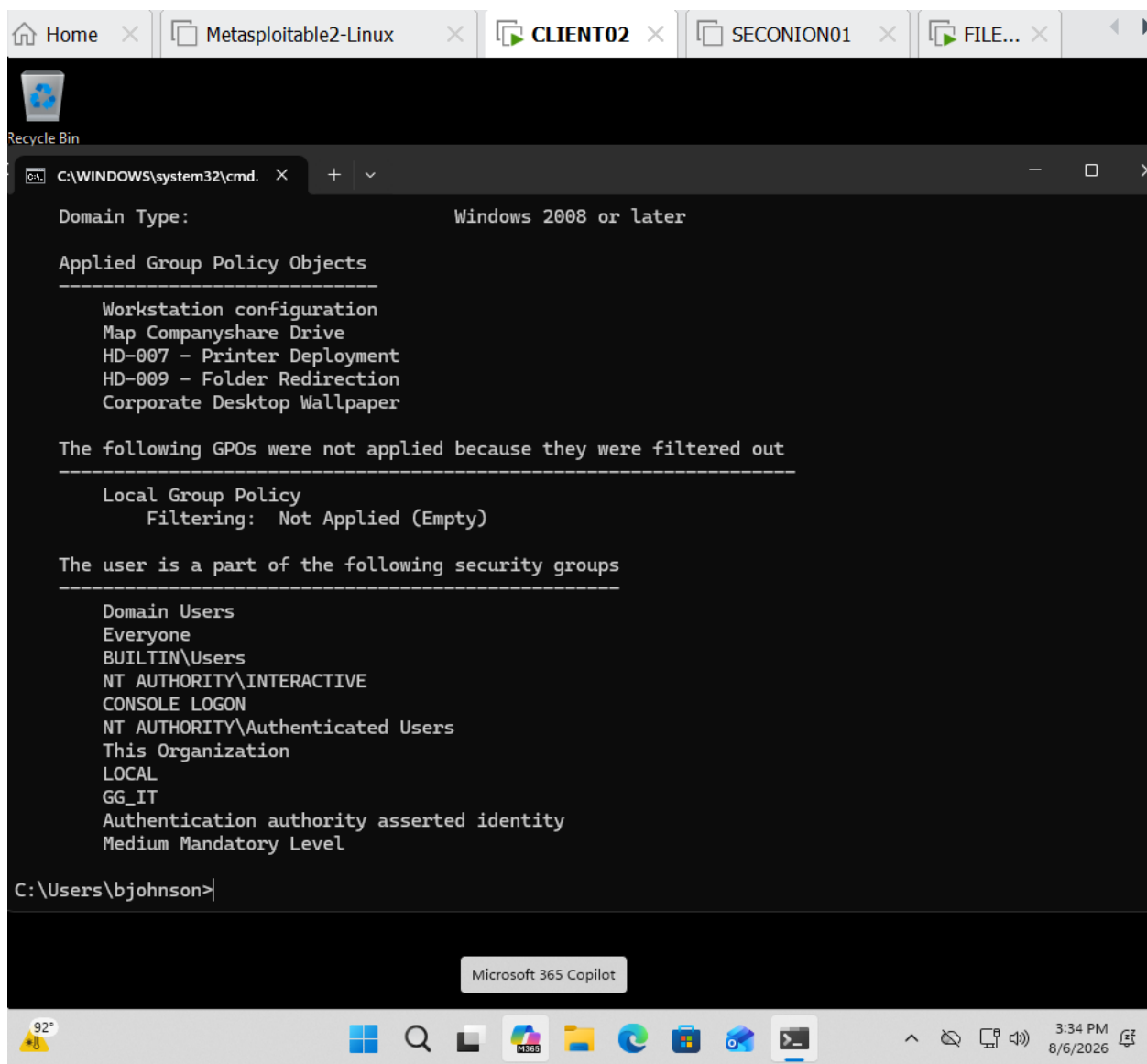
Applied Group Policy Objects
-----
Map Companyshare Drive
HD-007 - Printer Deployment
HD-009 - Folder Redirection
Corporate Desktop Wallpaper

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization

Search
```

Additional GPO link to the **Departments** OU.



```
Recycle Bin
C:\WINDOWS\system32\cmd.
Domain Type: Windows 2008 or later

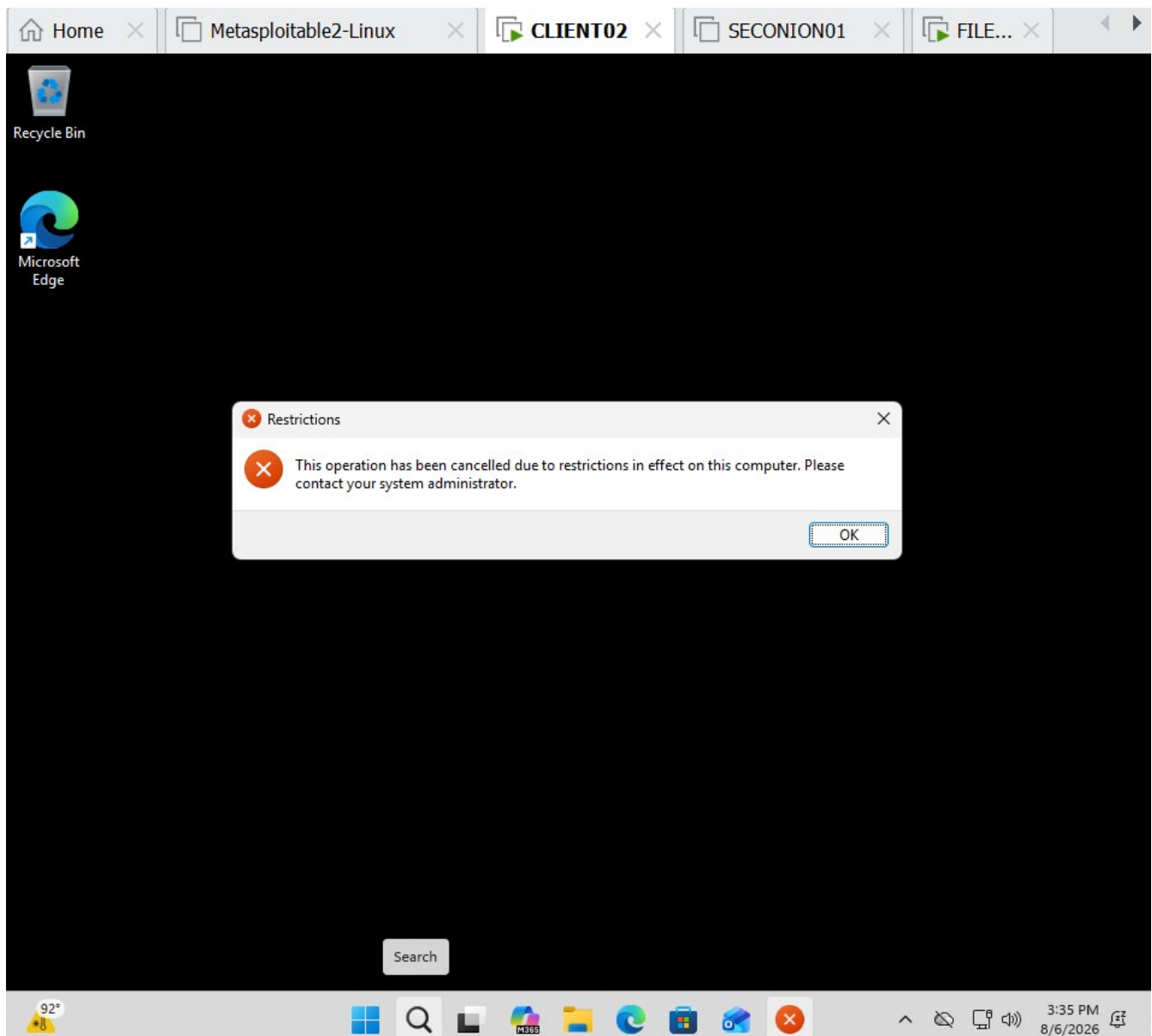
Applied Group Policy Objects
-----
Workstation configuration
Map Companyshare Drive
HD-007 - Printer Deployment
HD-009 - Folder Redirection
Corporate Desktop Wallpaper

The following GPOs were not applied because they were filtered out
-----
Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups
-----
Domain Users
Everyone
BUILTIN\Users
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
GG_IT
Authentication authority asserted identity
Medium Mandatory Level

C:\Users\bjohnson>
```

gpresult /scope user /r showing **Workstation Configuration** under Applied Group Policy Objects.



Restriction message:

"This operation has been cancelled due to restrictions in effect on this computer."

Task 6 - Configure Group Policy Loopback Processing

Objective

Configure and test **Group Policy Loopback Processing** to demonstrate how user configuration settings can be applied based on the **computer a user logs into**, rather than solely on the user's location within Active Directory.

Environment

- **Domain:** benlab.local
- **Domain Controller:** DC01
- **Client workstation:** CLIENT02
- **Test user:** Brian Johnson
- **User location:** Sales OU under Departments
- **Computer location:** Workstations OU
- **GPO:** Workstation configuration
- **Loopback mode:** Replace

Procedure

The existing **Workstation configuration** GPO was used for the loopback processing test. The GPO was linked to the **Workstations OU**, allowing policies to target computers located within that OU.

Within Group Policy Management Editor, the following location was opened:

Computer Configuration

→ Policies

→ Administrative Templates

→ System

→ Group Policy

The following policy was configured:

Configure user Group Policy loopback processing mode

The policy was set to:

Enabled

Mode: Replace

Replace mode was selected so that when a user logs onto a workstation affected by this GPO, the user configuration associated with the workstation's GPO context replaces the user's normal user-policy processing for that session.

Preparation for Testing

The temporary **Workstation configuration** GPO link previously created on the **Departments OU** during Task 5 was removed before testing loopback processing.

This was important because leaving that link in place would allow Brian Johnson to receive the GPO directly through his user OU hierarchy, making it difficult to prove that **Loopback Processing** was responsible for applying the policy.

After the link was removed, Control Panel access returned to normal, confirming that the previous policy application was no longer affecting Brian through the Departments OU.

Testing

Brian Johnson, whose Active Directory account resides within:

BEN Industries

→ Departments

→ Sales

logged onto **CLIENT02**, whose computer account resides within:

BEN Industries

→ Workstations

Group Policy was refreshed and the resulting user policy was examined using:

```
gpresult /scope user /r
```

The output showed:

Applied Group Policy Objects

Workstation configuration

Map Companyshare Drive

HD-007 - Printer Deployment

HD-009 - Folder Redirection

Corporate Desktop Wallpaper

The presence of **Workstation configuration** under Brian Johnson's applied user GPOs confirmed that the workstation-linked GPO was affecting his user session through **Group Policy Loopback Processing**.

Functional Verification

Control Panel was then opened while logged in as Brian Johnson.

Windows displayed the following restriction:

```
This operation has been cancelled due to restrictions  
in effect on this computer.
```

```
Please contact your system administrator.
```

This confirmed that the user restriction configured within the **Workstation configuration GPO** was successfully applied to Brian Johnson when he logged onto CLIENT02.

Additional Observation

The following command was also attempted while logged in as Brian Johnson:

```
gpresult /scope computer /r
```

The command returned:

```
ERROR: Access Denied.
```

This occurred because the command was being executed under the standard user's security context. Computer-scope Group Policy results can instead be queried from an **elevated Command Prompt** using administrative credentials.

This did not affect the validity of the loopback test because the user-scope `gpresult` output and the successful Control Panel restriction provided evidence that the policy was functioning.

Result

Task completed successfully.

Group Policy Loopback Processing was successfully configured using **Replace mode**. Brian Johnson normally receives policies according to his location in the Sales OU. However, when he logged onto CLIENT02 in the Workstations OU, the user settings associated with the workstation's GPO context were applied.

The Control Panel restriction provided functional confirmation that the loopback policy was operating as intended.

Key Learning Outcome

This task demonstrated the difference between normal Group Policy processing and loopback processing.

Normally:

User → User's OU → User policies

With Loopback Processing:

User

↓

Logs onto CLIENT02

↓

CLIENT02 is in Workstations OU

↓

Workstation configuration GPO

↓

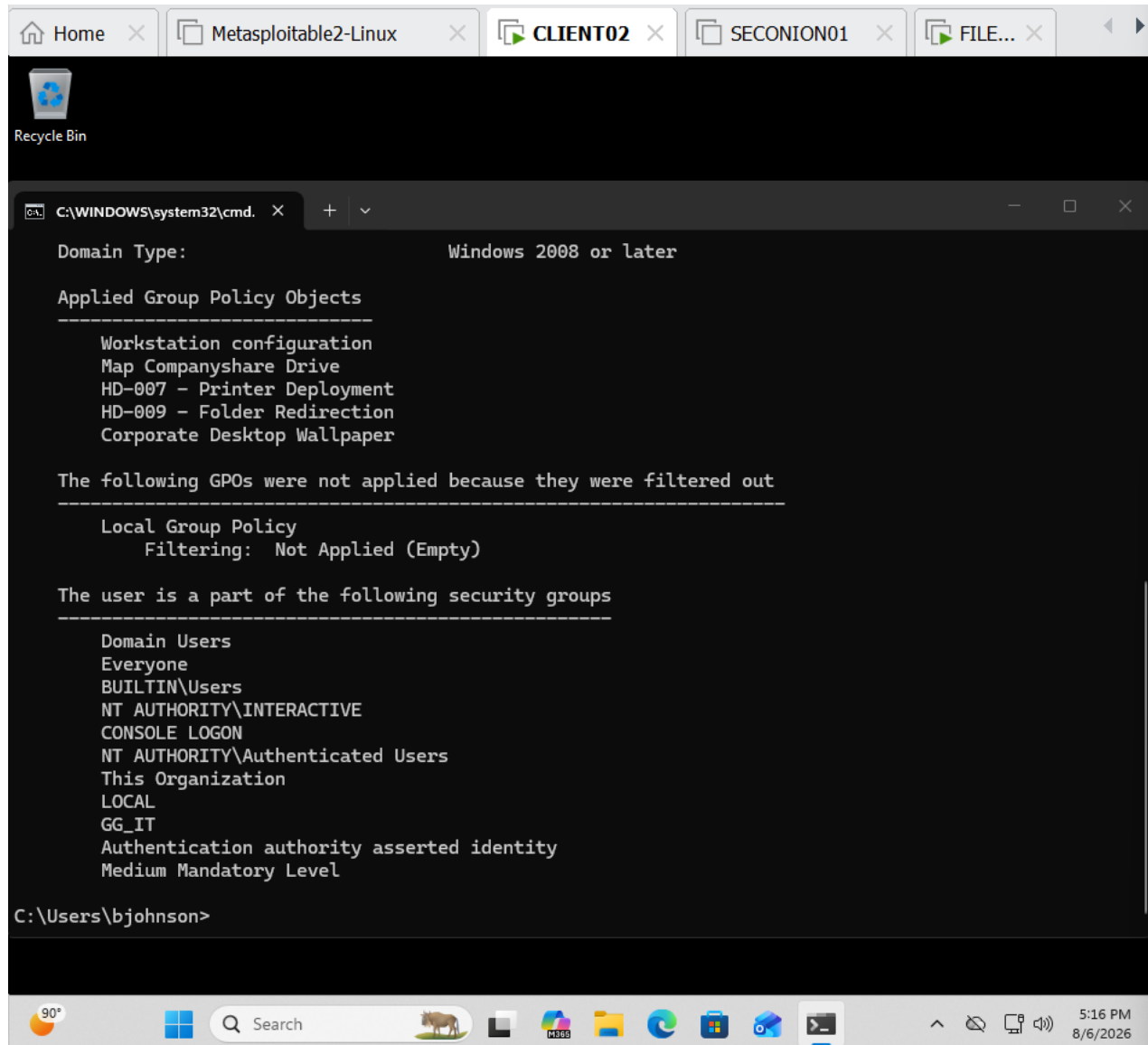
Loopback Processing (Replace)

↓

Workstation-context user policies applied

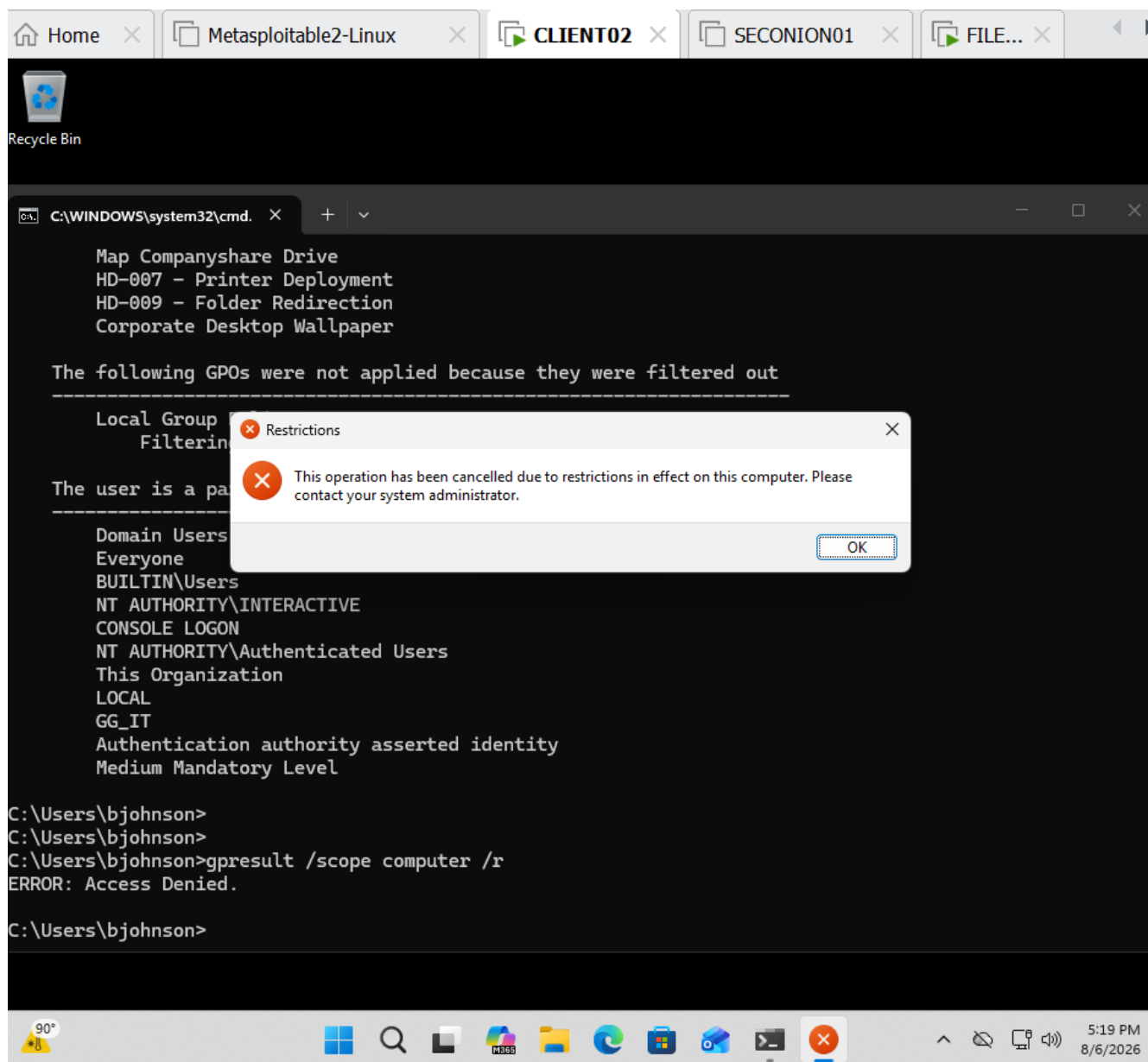
This type of configuration is useful in enterprise environments where administrators need a consistent user experience on specific computers regardless of who logs onto them, such as **kiosks, shared workstations, classrooms, training rooms, reception computers, and secured administrative systems.**

Screenshots



Screenshot 1 – Loopback Policy Verification

gpresult /scope user /r showing **Workstation configuration** under Applied Group Policy Objects while logged in as Brian Johnson.



Screenshot 2 – Functional Restriction Test

Control Panel attempt showing “This operation has been cancelled due to restrictions in effect on this computer.”

Task 7 – Configure and Test WMI Filtering

Objective

Configure a **Windows Management Instrumentation (WMI) filter** to control Group Policy application based on the operating system of the target computer.

The goal was to configure the **Corporate Security Settings** GPO so that it applies to qualifying **Windows 11 workstation computers**, while preventing the same GPO from applying to Windows Server systems.

Environment

Domain: benlab.local

Domain Controller: DC01

Group Policy Object: Corporate Security Settings

Target OU: Workstations

WMI Filter: Windows 11 Workstations

Windows 11 test system: CLIENT02

Windows Server test system: FILESRV01

Step 1 – Create the WMI Filter

On DC01, opened:

Server Manager → **Tools** → **Group Policy Management**

Expanded:

Forest: benlab.local → **Domains** → **benlab.local** → **WMI Filters**

Right-clicked **WMI Filters** and selected:

New

Created the following WMI filter:

Name:

Windows 11 Workstations

Description:

Applies Group Policy only to Windows 11 workstation computers.

Step 2 – Configure the WMI Query

Selected **Add** to create a query.

The default WMI namespace was retained:

root\CIMv2

The following WMI query was entered:

```
SELECT * FROM Win32_OperatingSystem WHERE Version LIKE "10.%" AND ProductType = "1"
AND BuildNumber >= "22000"
```

The query evaluates properties of the Win32_OperatingSystem class.

The important conditions are:

- `Version LIKE "10.%"` — matches the Windows NT 10.x version family used by Windows 10/11.
- `ProductType = "1"` — identifies Windows client/workstation operating systems rather than Windows Server.
- `BuildNumber >= "22000"` — identifies Windows 11-era builds.

The WMI filter was then saved.

Step 3 – Associate the WMI Filter with the GPO

Selected:

Group Policy Objects → Corporate Security Settings

On the **Scope** tab, located the **WMI Filtering** section.

Changed the WMI filter from:

<none>

to:

Windows 11 Workstations

Confirmed the association when prompted.

The Corporate Security Settings GPO was therefore configured to evaluate the WMI filter before being applied to a computer.

Step 4 – Verify the GPO on CLIENT02

On the Windows 11 client CLIENT02, Group Policy was refreshed.

The resulting computer policy was examined using:

```
gpresult /scope computer /r
```

The output showed:

Applied Group Policy Objects

Corporate Security Settings

Workstation configuration

Default Domain Policy

HD-008 - Software Deployment

This confirmed that **Corporate Security Settings successfully applied to CLIENT02**.

Because CLIENT02 satisfied the WMI query conditions, the WMI filter evaluated successfully and allowed the GPO to be processed.

Step 5 – Perform a Negative Test Using FILESRV01

To verify that the WMI filter actually controlled policy application rather than simply allowing the GPO everywhere, a temporary additional link for **Corporate Security Settings** was created on the **Servers OU**.

This allowed FILESRV01, running Windows Server 2022, to encounter the same GPO.

Group Policy was refreshed on FILESRV01 and the resulting computer policy was examined.

The applied GPOs included:

Default Domain Policy

HD-008 - Software Deployment

However:

Corporate Security Settings

was **not applied**.

This was the expected result.

FILESRV01 is a Windows Server system and therefore does not satisfy:

```
ProductType = "1"
```

Consequently, the WMI filter prevented the **Corporate Security Settings** GPO from applying to the server.

Step 6 – Generate Detailed Group Policy Results

For additional verification, a detailed HTML Group Policy report was generated on FILESRV01:

```
gpresult /h C:\gpresult.html
```

The report was opened with:

```
start C:\gpresult.html
```

The report confirmed successful Group Policy processing while showing that **Corporate Security Settings was not among the applied GPOs** on FILESRV01.

This provided additional evidence that the WMI filter was functioning as intended.

Step 7 – Remove the Temporary Server Link

After completing the negative test, the temporary **Corporate Security Settings** link was removed from the **Servers OU**.

Only the link was deleted; the actual GPO was **not deleted**.

The final configuration was verified under:

Group Policy Objects → Corporate Security Settings → Scope

The resulting configuration showed:

Link:

Workstations

Link Enabled:

Yes

Enforced:

No

Security Filtering:

Authenticated Users

WMI Filtering:

Windows 11 Workstations

This returned the environment to its intended production-style configuration.

Result

Status: Successful

The lab demonstrated that a GPO can be linked within the Active Directory hierarchy while a **WMI filter provides an additional dynamic condition determining whether the GPO actually applies to a computer.**

The test produced two different results:

System	Operating System	WMI Match	Corporate Security Settings
CLIENT02	Windows 11	Yes	Applied
FILESRV01	Windows Server 2022	No	Not Applied

This verified both the **positive** and **negative** behavior of the WMI filter.

Key Concepts Learned

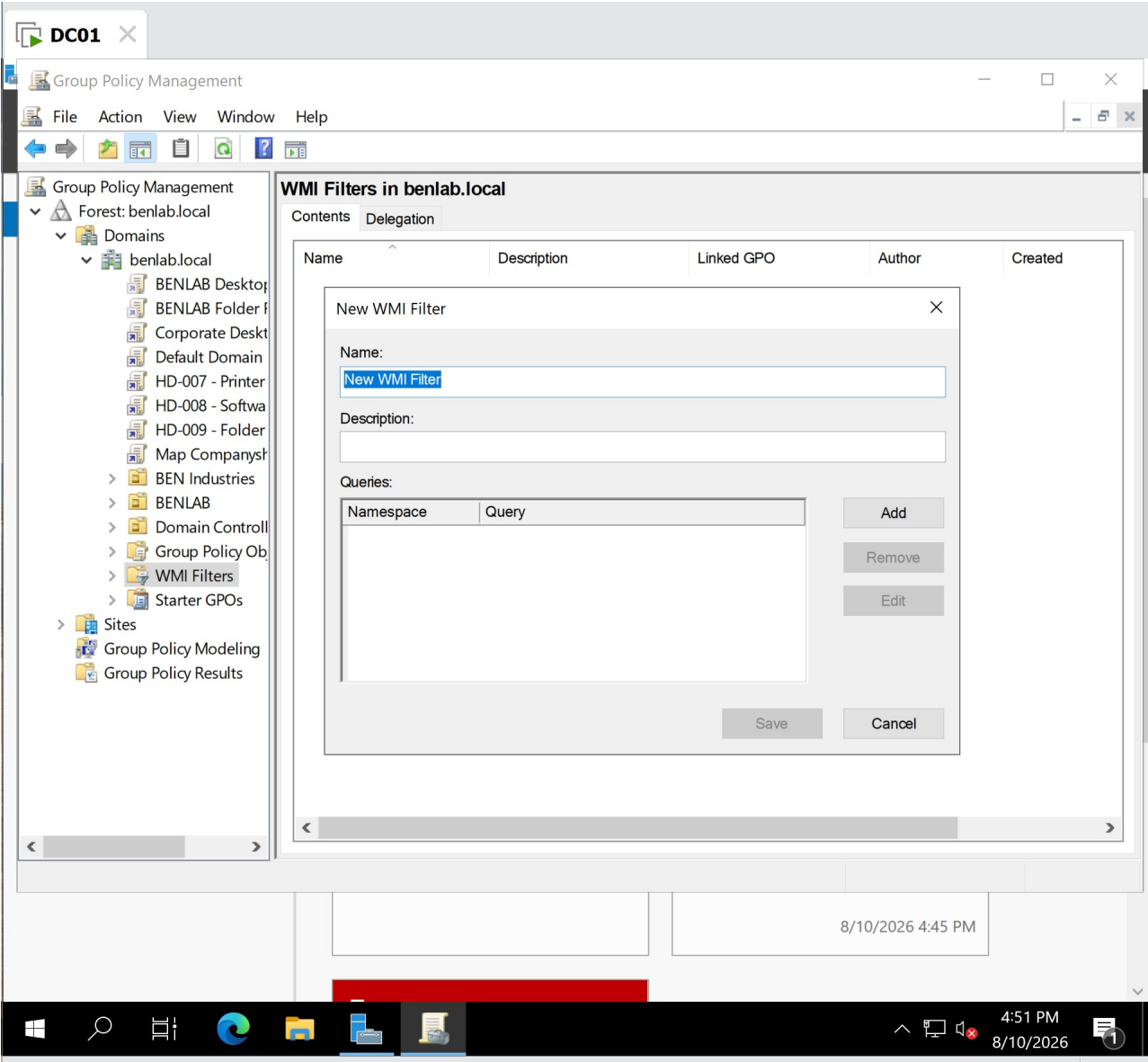
A GPO link determines where a policy is available for processing, while a WMI filter determines whether a particular computer meets specified conditions before that GPO is applied.

This provides administrators with more granular targeting than OU placement alone. For example, administrators can target Group Policy according to operating-system version, workstation/server classification, hardware characteristics, or other information available through WMI.

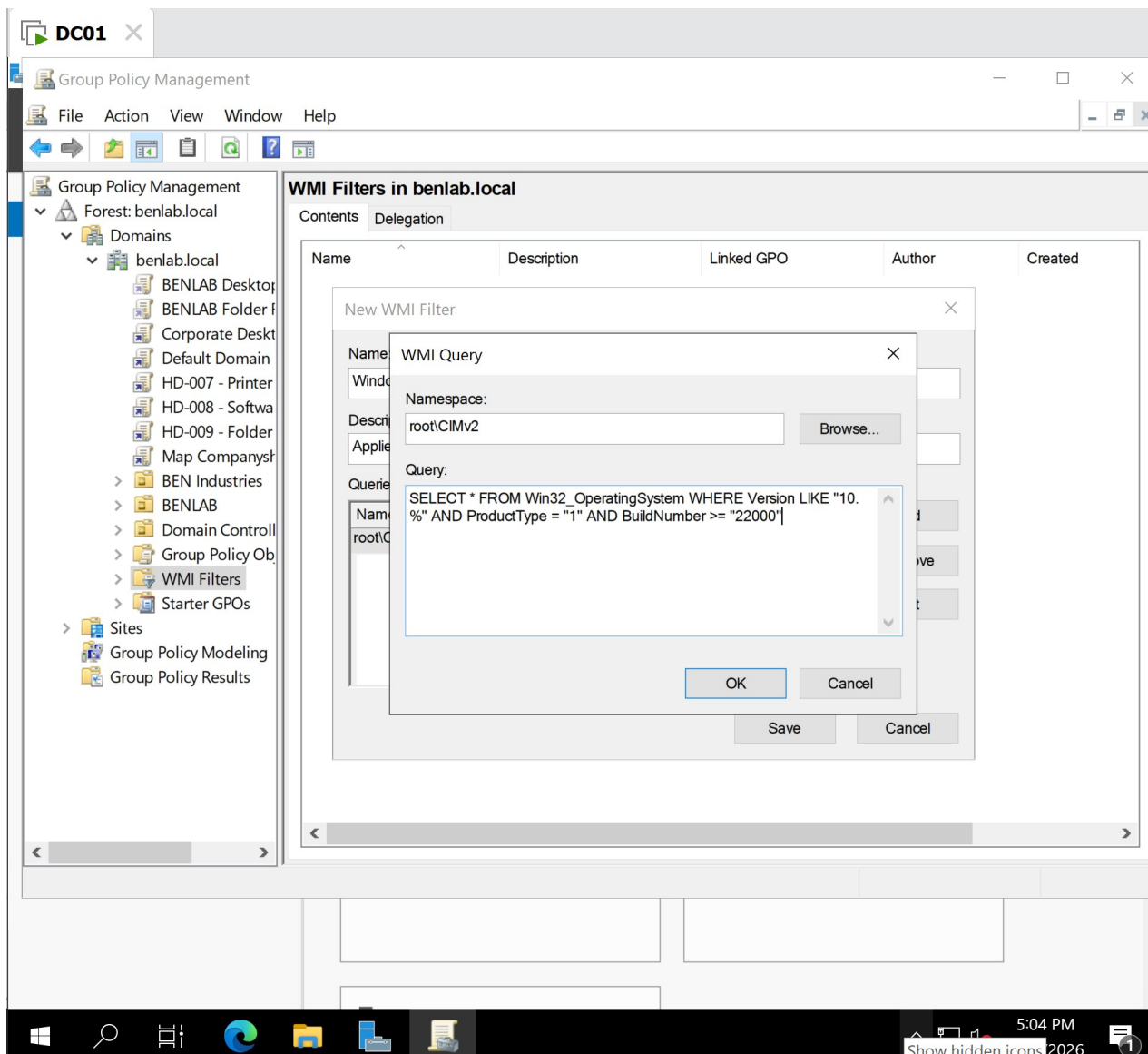
The lab also demonstrated an important troubleshooting principle:

A GPO being linked to an OU does not necessarily mean that every computer within that scope will receive it. Administrators must also consider security filtering, WMI filtering, inheritance, and policy processing results.

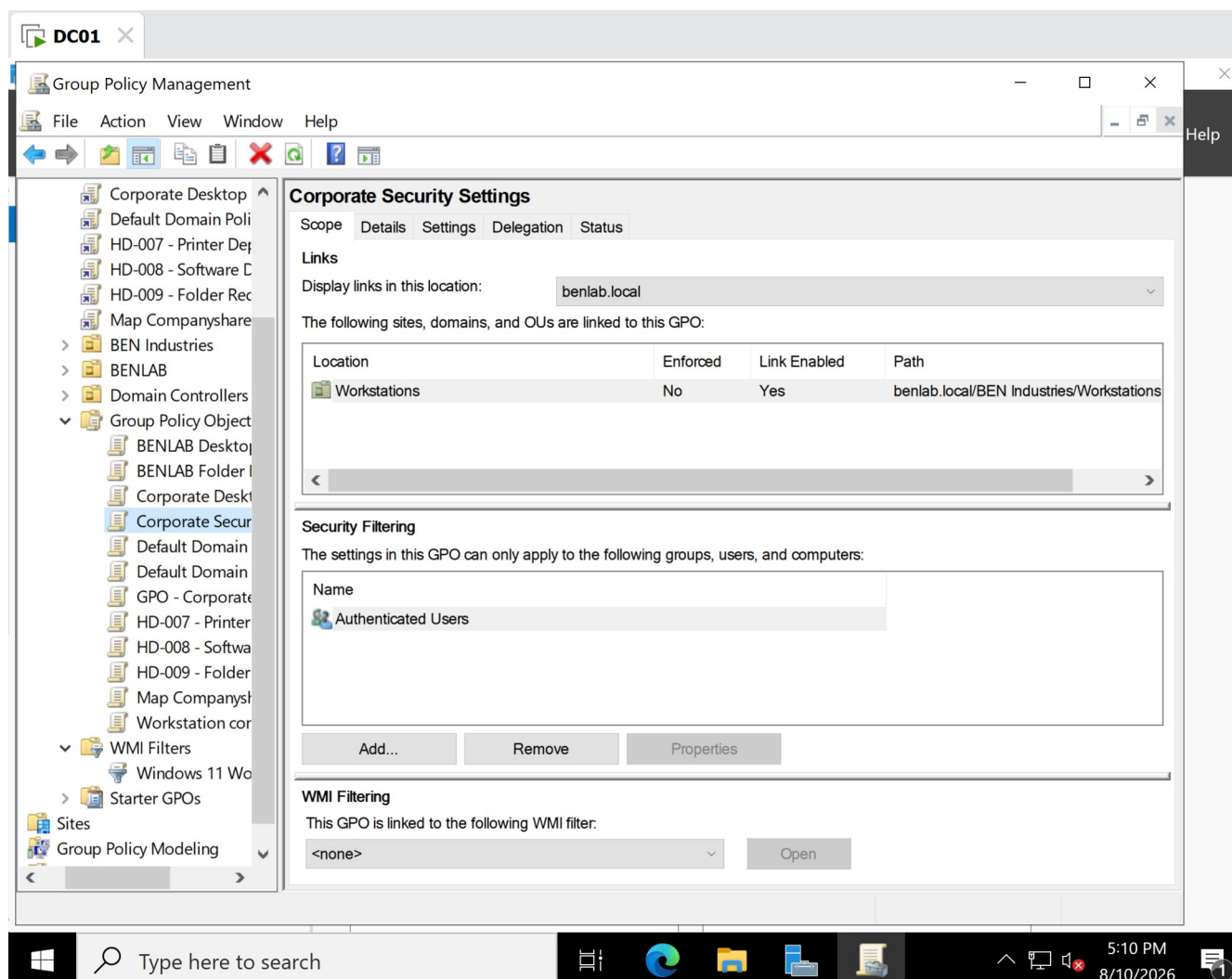
Screenshots



Creating Windows 11 Workstations WMI filter



The WMI query showing Win32_OperatingSystem, ProductType, and BuildNumber



Corporate Security Settings → WMI Filtering: Windows 11 Workstations

```

Home x Metasploitable2-Linux x CLIENT02 x SECONION01 x FILE... x
Administrator: Windows Pow x
Group Policy slow link threshold: 500 kbps
Domain Name: BENLAB
Domain Type: Windows 2008 or later

Applied Group Policy Objects
Corporate Security Settings
Workstation configuration
Default Domain Policy
HD-008 - Software Deployment

The following GPOs were not applied because they were filtered out
Local Group Policy
Filtering: Not Applied (Empty)

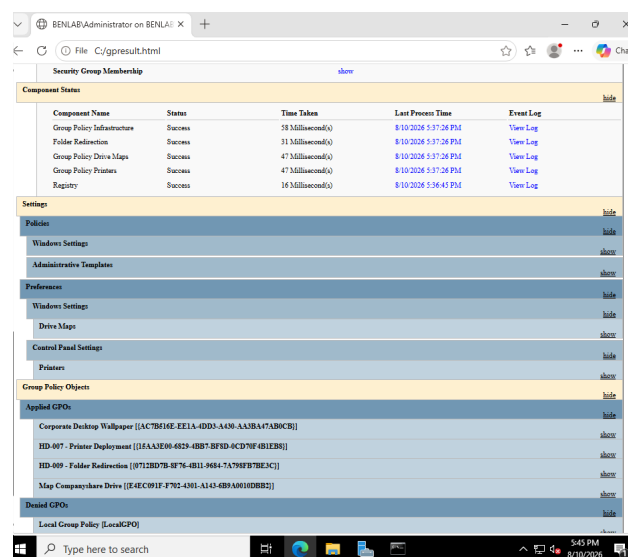
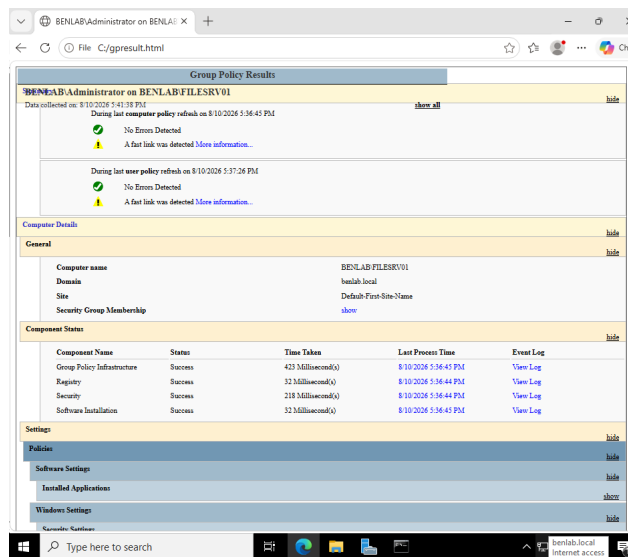
The computer is a part of the following security groups
BUILTIN\Administrators
Everyone
BUILTIN\Users
NT AUTHORITY\NETWORK
NT AUTHORITY\Authenticated Users
This Organization
DESKTOP-0U99MOA$
Domain Computers
Authentication authority asserted identity
System Mandatory Level

PS C:\Users\Administrator> |

```

CLIENT02
gpresult
showing
Corporate
Security
Settings
applied

The FILESRV01 gpresult/HTML report



Task 8 – Configure Security Filtering for the IT Department

Objective

Configure Group Policy security filtering so that a specific user-based policy targets members of the GG_IT security group within the appropriate organizational unit structure.

Configuration Performed

Created and configured the GPO:

GPO Name: Security Filtering - IT Department

The GPO was linked to:

BEN Industries

└─ Departments

└─ Information Technology

Security Filtering was configured for:

GG_IT

The following user policy was configured:

User Configuration

└─ Policies

└─ Administrative Templates

└─ Control Panel

└─ Prohibit access to Control Panel and PC settings

→ Enabled

Security Filtering Verification

The **GG_IT** group was verified to have the required GPO permissions:

Read Allow

Apply group policy Allow

No explicit **Deny** permission was configured for GG_IT.

Test User

The policy was tested using the domain user:

Daniel Scott

Active Directory verification confirmed that Daniel Scott is located in:

CN=Daniel Scott,

OU=Information Technology,

OU=Departments,

OU=BEN Industries,

DC=benlab,

DC=local

Daniel Scott was also confirmed as a member of:

GG_IT

Troubleshooting and Verification

During client-side testing on CLIENT02, the targeted GPO did not appear in the applied GPO list returned by gpresult.

The following checks were performed:

```
gpupdate /force
```

```
gpresult /r
```

```
gpresult /scope user /z
```

SYSVOL connectivity was also verified from CLIENT02:

```
dir \\benlab.local\SYSVOL
```

On DC01, PowerShell was used to verify the GPO and its Active Directory link:

```
Get-GPO -Name "Security Filtering - IT Department"
```

The GPO's AD and SYSVOL version information was examined using:

```
Get-GPOReport -Name "Security Filtering - IT Department" -ReportType XML -Path "C:\GPOReport.xml"
```

```
Select-String -Path "C:\GPOReport.xml" -Pattern "VersionDirectory","VersionSysvol"
```

The Departments OU's gPLink was verified with:

```
Get-ADOrganizationalUnit -Identity "OU=Departments,OU=BEN  
Industries,DC=benlab,DC=local" -Properties gPLink |
```

```
Select-Object -ExpandProperty gPLink
```

The returned GUID matched the **Security Filtering - IT Department** GPO.

Daniel Scott's OU placement and group membership were additionally verified using:

```
Get-ADUser "daniel.scott" -Properties DistinguishedName,MemberOf |
```

```
Select-Object DistinguishedName,MemberOf
```

Result

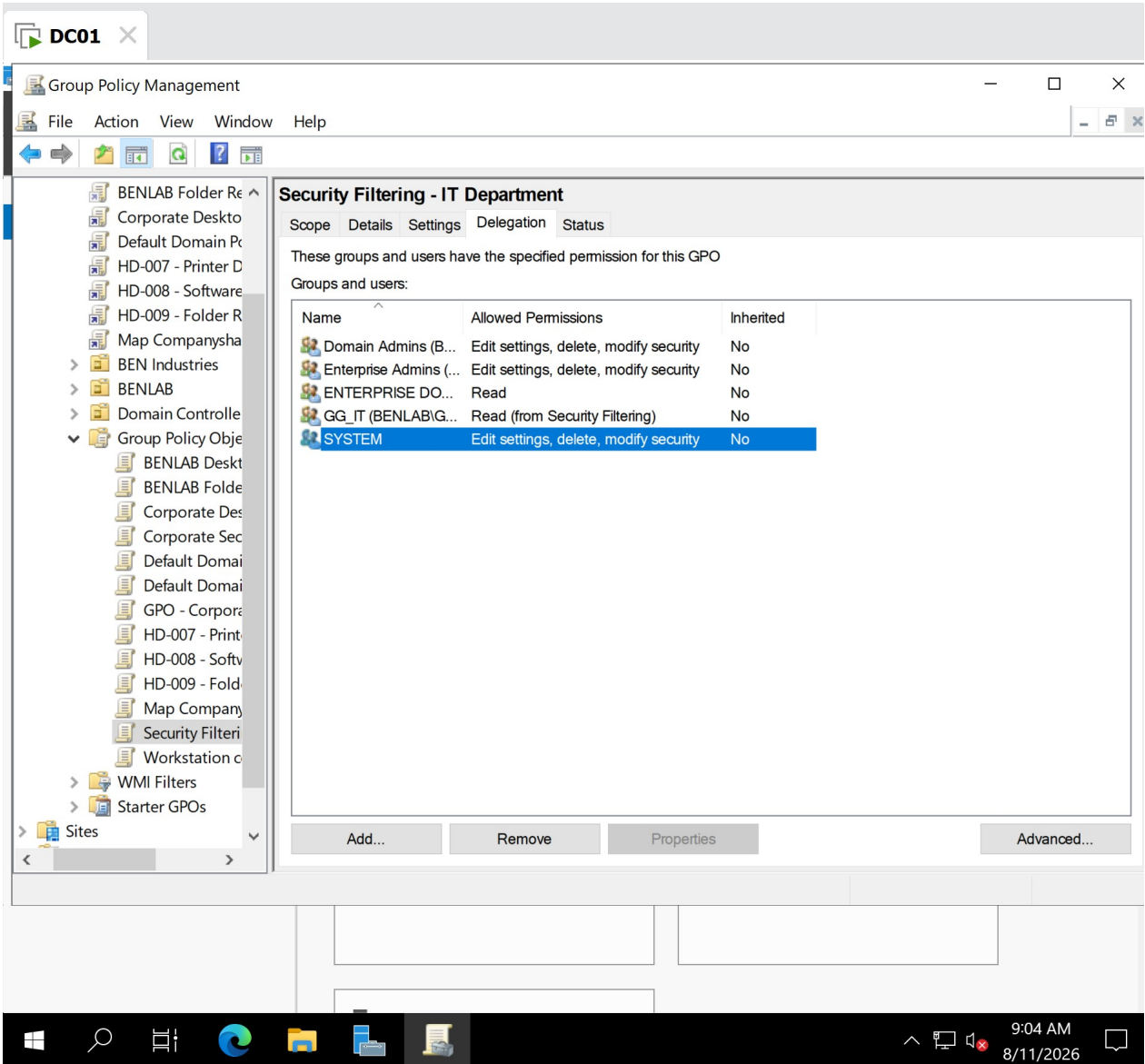
The server-side GPO configuration, OU link, security filtering, user placement, group membership, permissions, GPO version information, and SYSVOL accessibility were successfully validated.

However, the targeted GPO was **not enumerated by GPRresult on CLIENT02 during final client-side testing**. The behavior remained unresolved after systematic troubleshooting and was documented rather than making additional unnecessary configuration changes.

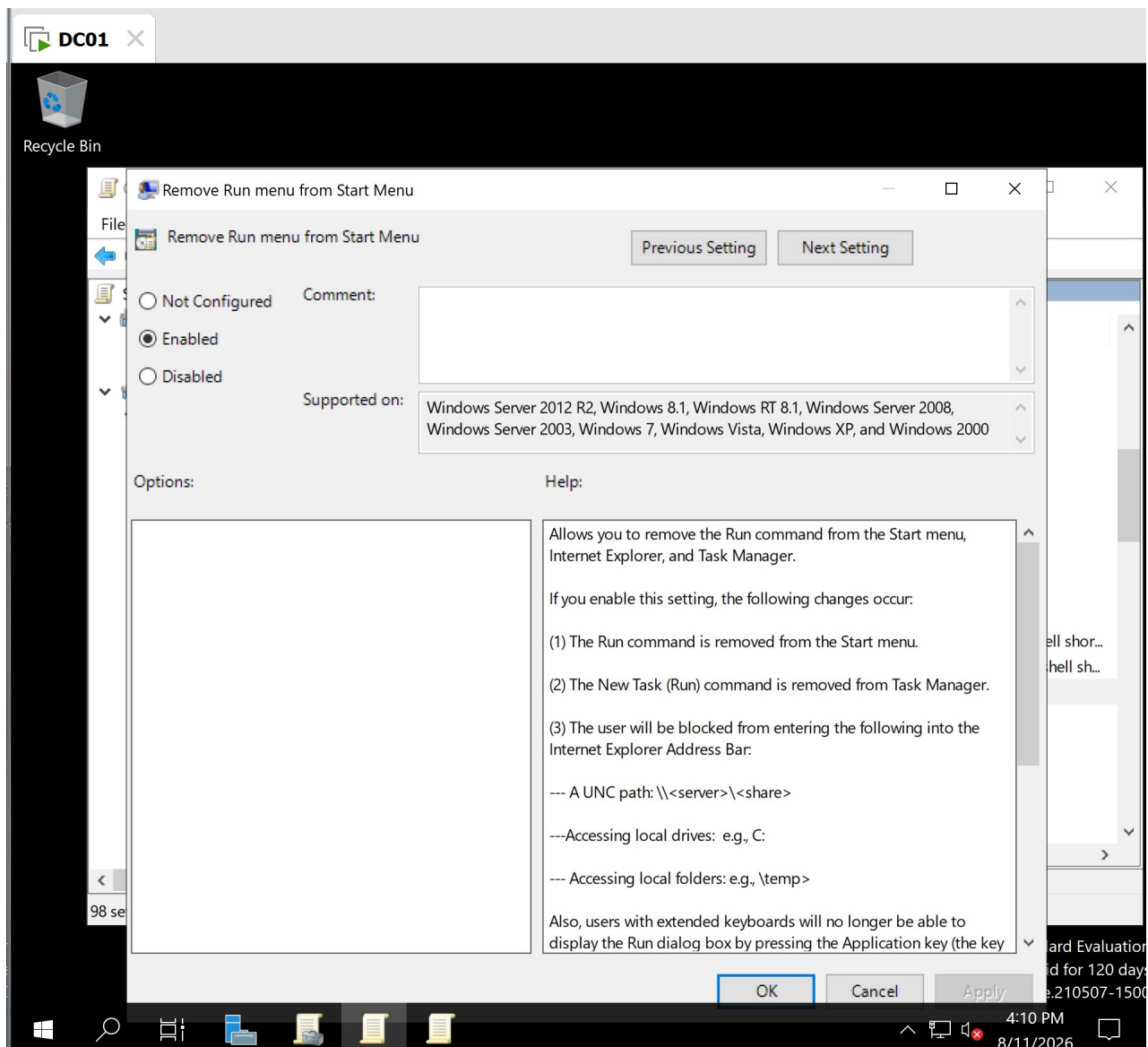
Skills Practiced

Group Policy Management • Security Filtering • Active Directory OU Structure • Security Groups • GPO Permissions • SYSVOL • GPResult • GPOupdate • PowerShell • GPO Troubleshooting

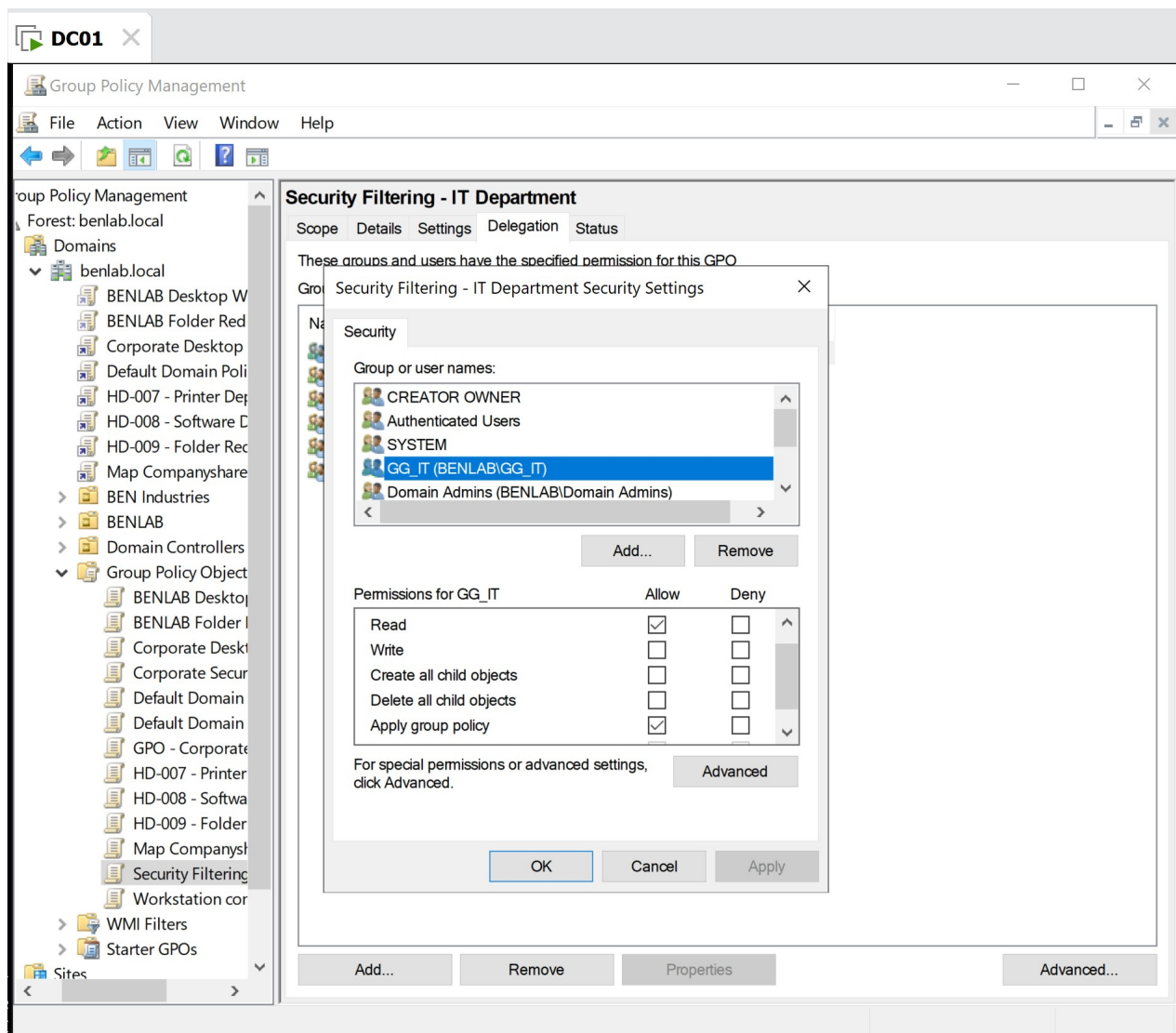
Screenshots



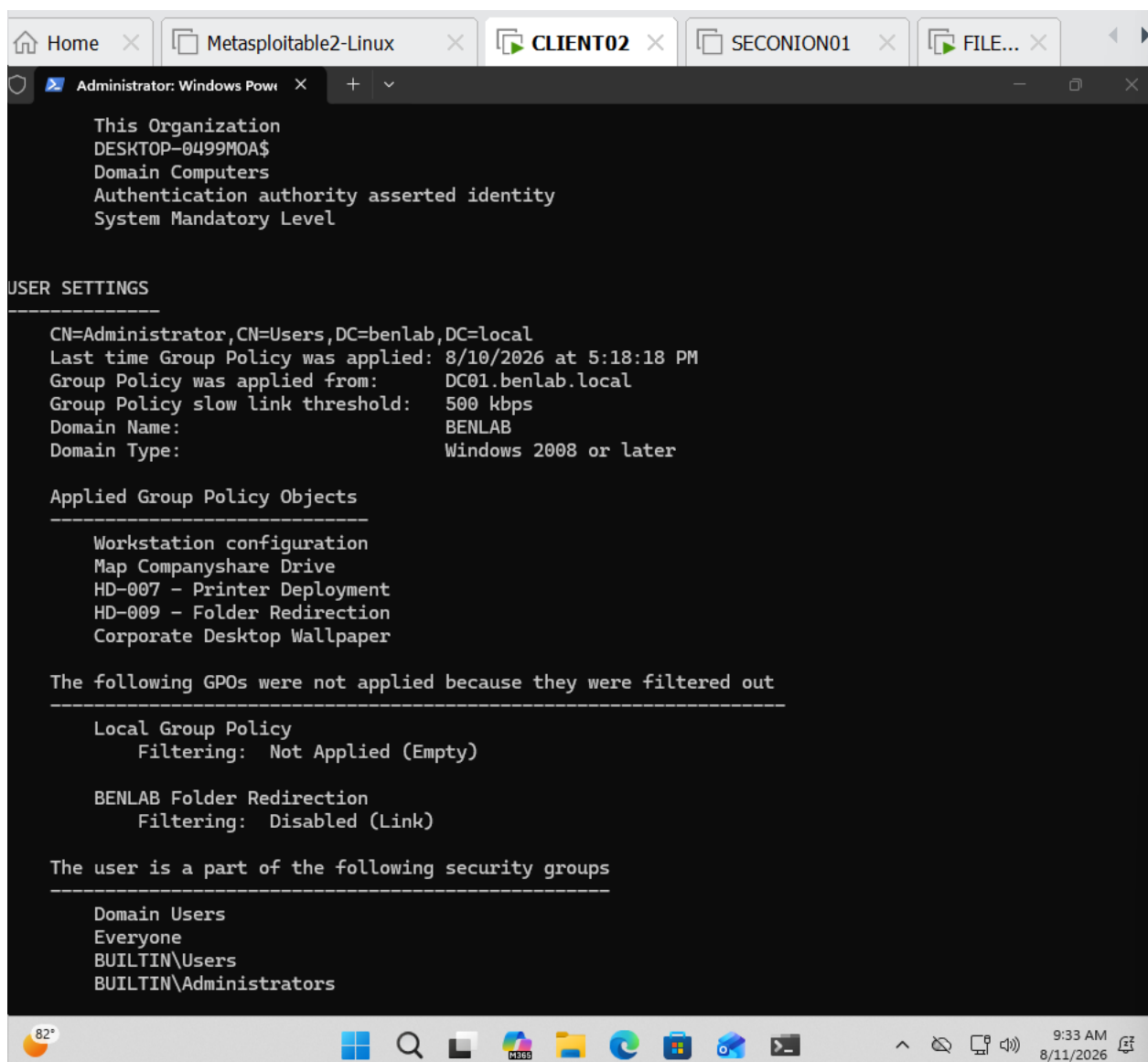
GPO Scope tab — showing GG_IT under Security Filtering.



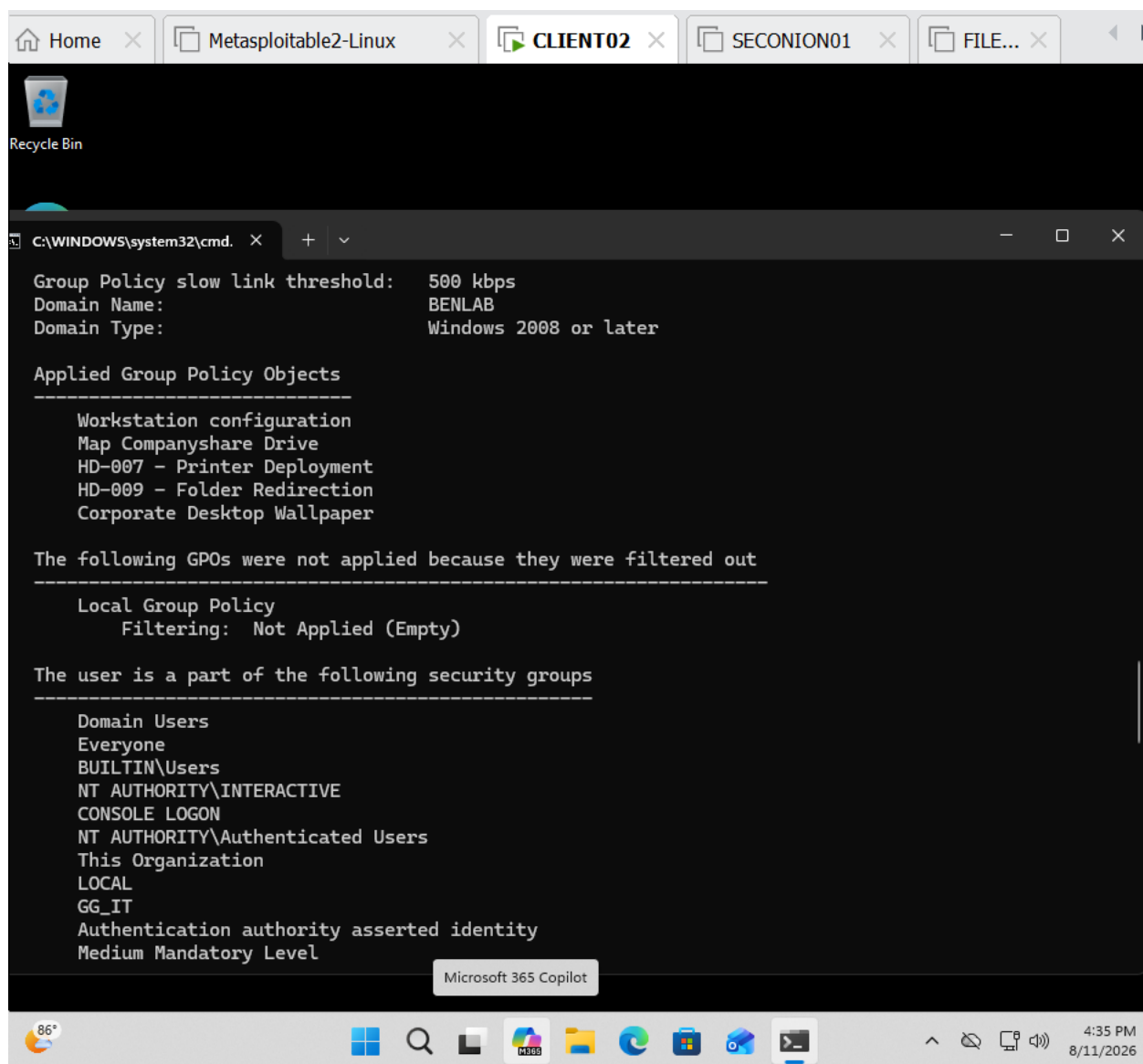
GPO Settings tab — showing Prohibit access to Control Panel and PC settings = Enabled.



Advanced Security screenshot — the one you just took showing GG_IT with **Read + Apply** group policy



CLIENT02 gpresult screenshot — showing Daniel's GG_IT membership and the absence of the targeted GPO.



DC01 PowerShell screenshot — your final one showing Daniel Scott's correct DistinguishedName and GG_IT membership.

Task 9 & Task 10 – GPO Backup and Restore

Objective

Demonstrate how to back up and restore a Group Policy Object (GPO) to provide a recovery mechanism for accidental configuration changes, corruption, or administrative errors.

Environment

- **Domain Controller:** DC01

- **Domain:** benlab.local
- **GPO:** Corporate Security Settings
- **Backup Location:** C:\GPO-Backups\Corporate-Security-Settings
- **Management Tool:** Group Policy Management Console (GPMC)

Task 9 – Back Up the GPO

The Corporate Security Settings GPO was backed up before performing restore testing.

Procedure:

1. Opened **Group Policy Management** on DC01.
2. Expanded **Domains** → **benlab.local** → **Group Policy Objects**.
3. Right-clicked **Corporate Security Settings** and selected **Back Up**.
4. Specified the backup location:
C:\GPO-Backups\Corporate-Security-Settings
5. Added the description:
Task 9 - Backup before GPO restore testing
6. Started the backup.
7. GPMC reported that the backup **Succeeded**.
8. Verified the backup directory in File Explorer and confirmed that Windows created the corresponding backup folder.

Result:

The Corporate Security Settings GPO was successfully backed up and stored on DC01.

Task 10 – Restore the GPO

The previously created backup was then used to test the GPO recovery process.

Procedure:

1. Returned to **Group Policy Management**.
2. Right-clicked **Corporate Security Settings** and selected **Restore from Backup**.
3. Selected:
C:\GPO-Backups\Corporate-Security-Settings
4. Selected the backup created during Task 9.
5. Verified the backup timestamp and description in the Restore Group Policy Object Wizard.
6. Completed the restore operation.

7. Returned to the **Settings** tab of Corporate Security Settings to verify the restored configuration.

Post-Restore Verification:

The GPO remained enabled and its configuration was successfully displayed after restoration.

Verification included the GPO link, security filtering, WMI filtering, and configured computer security policies.

Result:

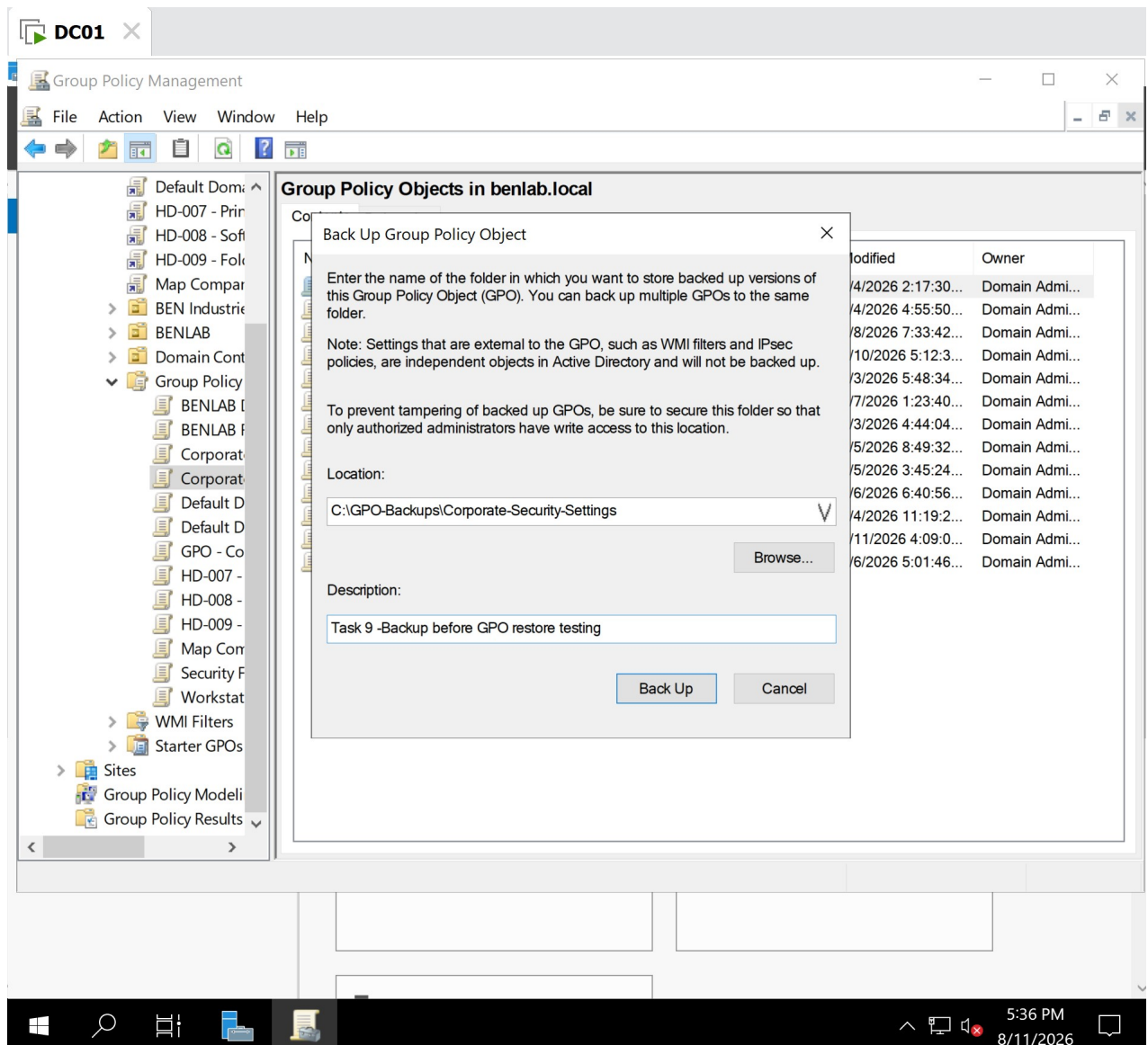
The Corporate Security Settings GPO was successfully recovered from its backup and its configuration was verified after restoration.

Key Learning

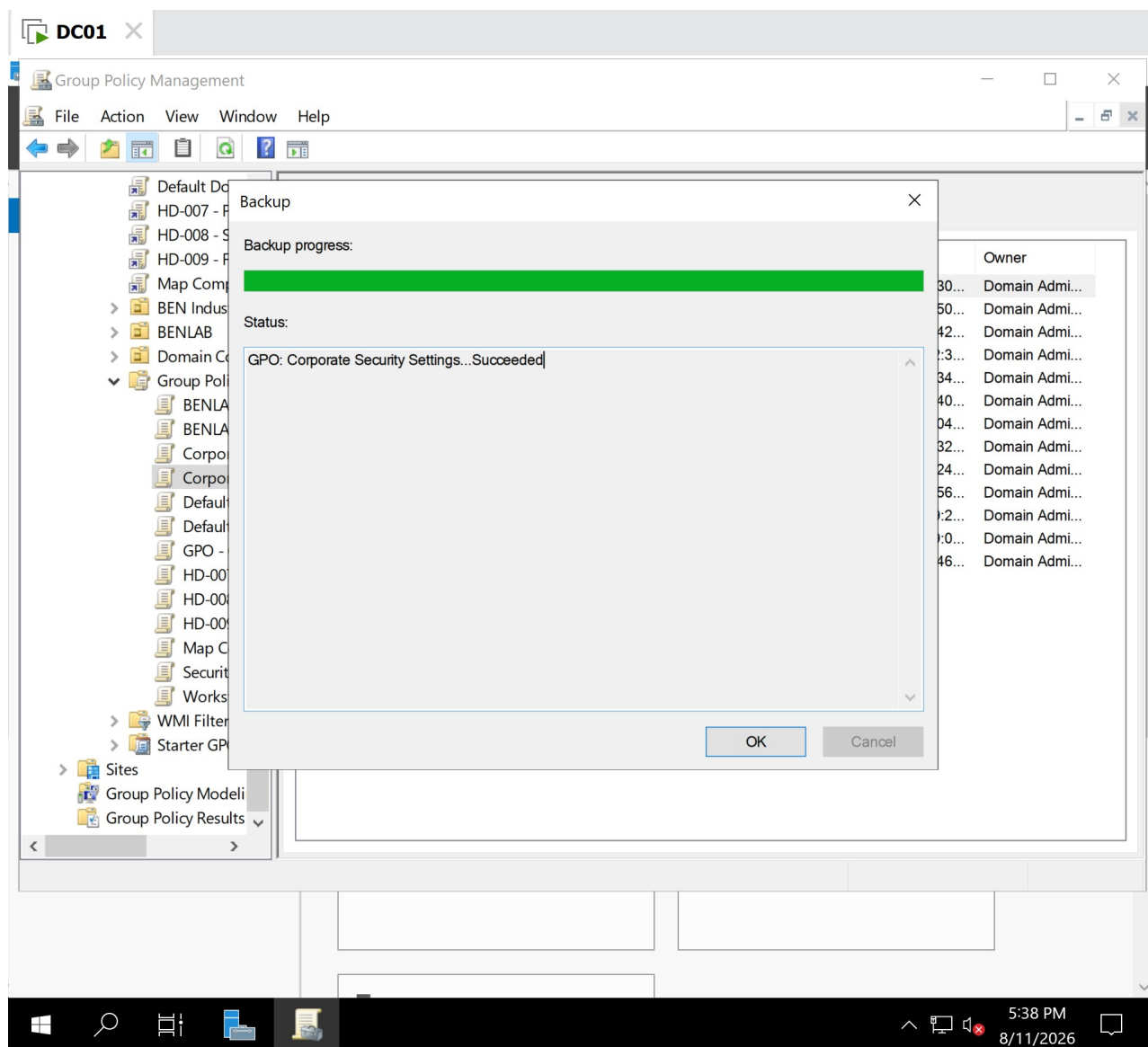
GPO backup and restore provides administrators with a recovery mechanism before making significant policy changes. A backup should be created before potentially disruptive modifications so that a known-good configuration can be restored if necessary.

One particularly important thing I learned is that **backup is only half of the recovery process**. An administrator should also know how to locate the correct backup, perform the restoration, and verify the GPO afterward.

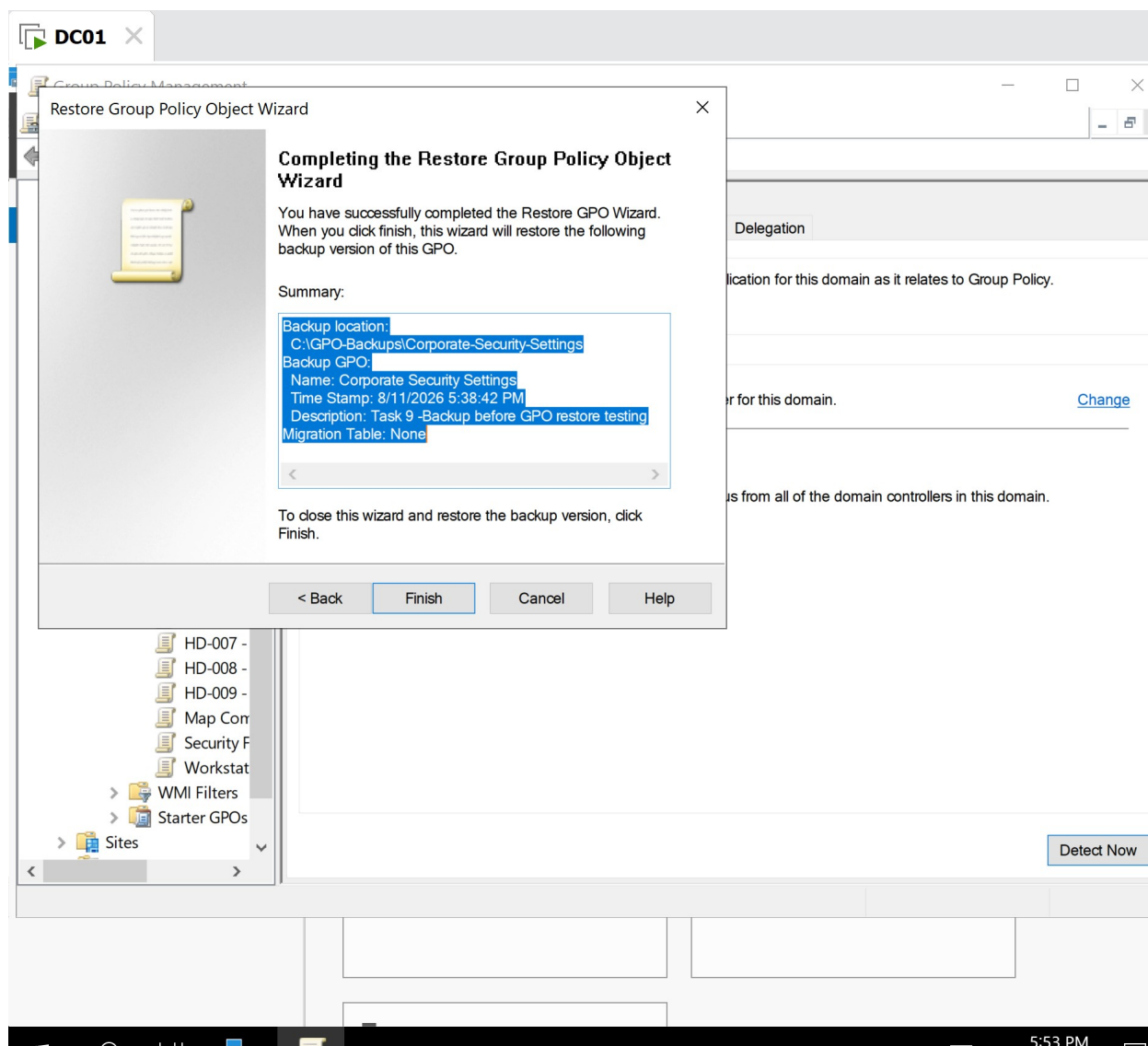
Screenshots



Backup configuration — showing the backup path and Task 9 description.



Backup succeeded — your 5:38 PM success screen.



Restore confirmation — the screen showing the correct GPO, backup location, timestamp, and description before clicking Finish.

DC01 - VMware Workstation

File Edit View VM Help

DC01

Group Policy Management

File Action View Window Help

Policy Management

st: benlab.local

Domains

benlab.local

BENLAB Desktop Wallpaper

BENLAB Folder Redirection

Corporate Desktop Wallpaper

Default Domain Policy

HD-007 - Printer Deployer

HD-008 - Software Deploy

HD-009 - Folder Redirection

Map Companyshare Drive

BEN Industries

BENLAB

Domain Controllers

Group Policy Objects

BENLAB Desktop Wallpa

BENLAB Folder Redirect

Corporate Desktop Wally

Corporate Security Setti

Default Domain Controll

Default Domain Policy

GPO - Corporate desktop

HD-007 - Printer Deploy

HD-008 - Software Deplc

HD-009 - Folder Redirect

Map Companyshare Driv

Security Filtering - IT De

Workstation configurati

WMI Filters

Starter GPOs

Sites

Group Policy Modeling

Group Policy Results

Corporate Security Settings

Scope Details Settings Delegation Status

Corporate Security Settings

Data collected on: 8/11/2025 5:57:01 PM

General

Details

Domain: benlab.local

Owner: BENLAB\Domain Admins

Created: 8/3/2025 5:20:20 PM

Modified: 8/11/2025 5:54:26 PM

User Revisions: 1 (AD), 1 (SYSVOL)

Computer Revisions: 45 (AD), 45 (SYSVOL)

Unique ID: {E5702E28-14E9-49DD-BE11-3EFEABBE8448}

GPO Status: Enabled

Links

Location: Workstations

Enforced: No

Link Status: Enabled

Path: benlab.local/BEN Industries/Workstations

This list only includes links in the domain of the GPO.

Security Filtering

The settings in this GPO can only apply to the following groups, users, and computers:

Name: NT AUTHORITY\Authenticated Users

WMI Filtering

WMI Filter Name: Windows 11 Workstations

Description: Applies Group Policy only to Windows 11 client workstations.

Delegation

These groups and users have the specified permission for this GPO

Name	Allowed Permissions	Inherited
BENLAB\Domain Admins	Edit settings, delete, modify security	No
BENLAB\Enterprise Admins	Edit settings, delete, modify security	No
NT AUTHORITY\Authenticated Users	Read from Security Filtering	No
NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS	Read	No
NT AUTHORITY\SYSTEM	Edit settings, delete, modify security	No

Computer Configuration (Enabled)

Policies

Windows Settings

Security Settings

Account Policies/Password Policy

Policy	Setting
Enforce password history	24 passwords remembered
Maximum password age	90 days
Minimum password age	1 day
Minimum password length	12 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

Account Policies/Account Lockout Policy

Policy	Setting
Account lockout duration	15 minutes
Account lockout threshold	5 invalid login attempts
Reset account lockout counter after	15 minutes

Software Restriction Policies

Enforcement

Policy	Setting
Apply Software Restriction Policies to the following	All software files except libraries (such as DLLs)
Apply Software Restriction Policies to the following users	All users
When applying Software Restriction Policies	Ignore certificate rules

Designated File Types

File Extension	File Type
ADE	ADE File
ADP	ADP File
BAS	BAS File
RAT	Windows Runtime File

Windows 11 Workstations

Network Internet access

Post-restore verification — the latest Corporate Security Settings report you just captured.